

APM 461 LECTURE NOTES

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1. MOTIVATING QUESTION

Question 1.1. Given some $n \in \mathbb{N}, \alpha, \beta \in [0, 1]$, what is the maximal m such that there exists subsets $A_1, \dots, A_m \subseteq [n]$ such that

- $\forall i, |A_i| \geq \alpha n$
- $\forall i \neq j, |A_i \cap A_j| \leq \beta n$

Typically, after fixing some value of α, β , we cannot get an exact formula for m in terms of n . However, we can ask how m grows with respect to n , e.g. $m \in \mathcal{O}(n^2), m \in \mathcal{O}(2^n)$, etc.

We will consider some examples

Example 1.2.

1. If $\alpha = 0.8, \beta = 0.1$, we get $m = 1$.
2. If $\alpha = 0.2, \beta = 0.2$, we get $m = 2^{\Theta(n)}$, i.e. m grows exponentially in n
3. If $\alpha = 0.5, \beta = 0.25$, we get $m = \Theta(n)$, we only state this result here.

Proof.

1. Trivial.
2. We want to m as large as possible. Let $m' = \binom{n}{0.2n}$, we can select $A_1, A_2, \dots, A_{m'}$ to be the all the distinct subsets with size $0.2n$, we have for all $i \neq j, |A_i \cap A_j| < 0.2n$ (they differ at least one value). Hence, m has a lowerbound of $\binom{n}{0.2n}$. We will show this lowerbound grows exponentially in n . This uses the following approximation

Lemma 1.3. (Sterling Approximation) $n! = \Theta\left(\sqrt{n}\left(\frac{n}{e}\right)^n\right)$

Proof. We have

$$\log(n!) = \log(1) + \dots + \log(n)$$

By integral approximation

$$\int_1^n \log(x) dx \approx \sum_{i=1}^{n-1} \frac{1}{2}(\log(i) + \log(i+1)) = \log(1) + \dots + \log(n) - \frac{1}{2}\log(n)$$

This gives

$$\begin{aligned} \log(n!) &\approx \int_1^n \log(x) dx + \frac{1}{2}\log(n) \\ &= x \log(x) - x \Big|_1^n + \frac{1}{2}\log(n) \\ &= n \log(n) - n + 1 + \frac{1}{2}\log(n) \\ \Rightarrow n! &\approx e^{n \log(n) - n + 1 + \frac{1}{2}\log(n)} = e \sqrt{n} \left(\frac{n}{e}\right)^n = \Theta\left(\sqrt{n}\left(\frac{n}{e}\right)^n\right) \end{aligned}$$

□

Now, using the Sterling Approximation, we get

$$\begin{aligned}
\binom{n}{0.2n} &= \frac{n!}{(0.2n)!(0.8n)!} \\
&= \Theta\left(\frac{\sqrt{n}\left(\frac{n}{e}\right)^n}{\sqrt{0.2n}\left(\frac{0.2n}{e}\right)^{0.2n}\sqrt{0.8n}\left(\frac{0.8n}{e}\right)^{0.8n}}\right) \\
&= \Theta\left(\frac{1}{\sqrt{n}}\left(\frac{1}{0.2^{0.2}0.8^{0.8}}\right)^n\right) \\
&= 2^{\Theta(n)}
\end{aligned}$$

which grows exponentially in n . In particular, we also know that the base only depends on the constant 0.2 and is not a function of n .

Finally, since there are at most 2^n subsets of $[n]$, we have m has an upperbound of $2^n = 2^{\Theta(n)}$, which also grows exponentially in n . Therefore, $m = 2^{\Theta(n)}$. $\square$

Remark 1.4. The above problem is an example of extremal combinatorics.

2. GRAPHS

2.1. Matchings.

In this section, we study the problem of finding perfect matchings in bipartite graphs.

Definition 2.1. (Bipartite Graph) A graph $G = (V, E)$ is **bipartite** if there exists $L, R \subseteq V$, such that $L \sqcup R = V$ and $E \subseteq L \times R$.

For simplicity, we denote such a graph by $G = (L, R, E)$.

In other words, a graph G is bipartite if its vertex set can be partitioned into a left vertex set and a right vertex set, and every edge connects a vertex from the left set to a vertex from the right set.

Definition 2.2. (Matching) Let $G = (L, R, E)$ be a bipartite graph, then $M \subseteq E$ is a **matching** for G if for all distinct edges $(u, v), (w, z) \in M, u \neq w$ and $v \neq z$.

In other words, a matching is a subset of edges such that no two distinct edges in M share a vertex. Thus, each vertex can be matched to at most one other vertex. In the context of bipartite graphs, since L and R are disjoint, it suffices to check that no two edges in M share the same left vertex or the same right vertex.

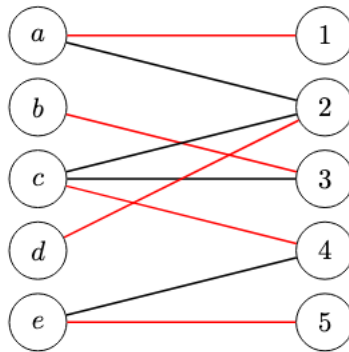
In this section, we focus on the case where $|L| = |R|$. In this setting, we can define the notion of a perfect matching, which is a matching in which every vertex is matched to exactly one other vertex.

2.1.1. Perfect Matching.

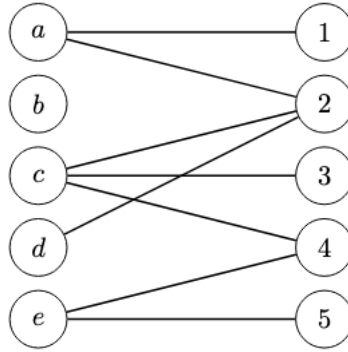
Definition 2.3. (Perfect Matching) Let $G = (L, R, E)$ where $|L| = |R| = m$ for some m , then $M \subseteq E$ is a **perfect matching** if M is a matching and $|M| = m$.

Example 2.4.

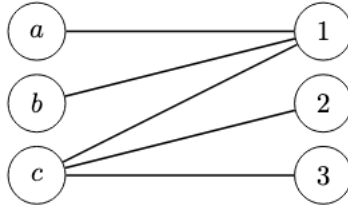
1. The following bipartite graph have a perfect matching that is coloured in red



1. Consider the same bipartite graph as in part (1). Suppose we remove all edges incident to vertex b . Then the resulting graph does not have a perfect matching, since vertex b has no adjacent edges.



1. Consider the following bipartite graph. This graph does not have a perfect matching because both a and b can only be matched to vertex 1, and at most one vertex can be matched to 1.



This leads us to the following natural question.

Question 2.5. When does a bipartite graph G have a perfect matching?

It turns out that Hall's Theorem provides an answer to this question.

2.1.2. *Hall's Theorem.*

Theorem 2.6. (Hall's Theorem) *Consider a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$, G has a perfect matching if and only if for all $S \subseteq L$, $|N_G(S)| \geq |S|$, where*

$$N_G(S) = \{v \in R : \exists u \in S \text{ s.t. } (u, v) \in E\}$$

Proof. ($\implies$) : Suppose G has a perfect matching M . Take any $S \subseteq L$, we can write down the vertices in this set as $S = \{s_1, \dots, s_m\}$ for some $m \leq n$. Since M is a perfect matching there are vertices $t_1, \dots, t_m \in R$ such that $(s_i, t_i) \in M$ for all i . Hence, $\{t_1, \dots, t_m\} \subseteq N_G(S) \implies |N_G(S)| \geq m = |S|$.

($\impliedby$) : We prove by induction on n .

Base case: We have $|L| = |R| = 1$. Take $S = L$ and we have $|N(S)| \geq |S| = 1 \implies N(S) = R$. Hence, there is an edge between the only vertex in L and the only vertex in R . This edge gives a perfect matching.

Inductive step: Suppose the claim holds for n , we will now prove that the claim holds for $n + 1$. We will argue in two cases.

Case 1: Suppose for all $S \subseteq L$, $0 < |S| < n + 1$, $|N_G(S)| \geq |S| + 1$.

Take any edge $(u, v) \in E$. Define the graph $G' = (L', R', E')$ be the subgraph of G induced by $L \setminus \{u\}$, $R \setminus \{v\}$, namely

$$\begin{cases} L' = L \setminus \{u\} \\ R' = R \setminus \{v\} \\ E' = E \cap (L' \times R') \end{cases}$$

Certainly, we have $|L \setminus \{u\}| = |R \setminus \{v\}| = n$. Moreover, for all $S \subseteq L \setminus \{u\}$, $|N_G(S)| \geq |S|$ as we only removed one vertex v from R .

By inductive hypothesis, G' has a perfect matching M' . Define $M = M' \cup \{(u, v)\}$, this gives us a perfect matching for G .

Case 2: Suppose there is some $S \subseteq L$, $0 < |S| < n + 1$, $|N_G(S)| = |S|$.

Define the graph $G' = (L', R', E')$ be the subgraph of G induced by $S, N_G(S)$, namely

$$\begin{cases} L' = S \\ R' = N_G(S) \\ E' = E \cap (L' \times R') \end{cases}$$

We show that G' has a perfect matching.

Given any subset $S^* \subseteq S$ we have $N_{G'}(S^*) \subseteq N_G(S) \subseteq R$. This gives $|N_{G'}(S^*)| = |N_G(S^*)| \geq |S^*|$. Since $|S| = |N_G(S)| < n + 1$, by inductive hypothesis G' has a perfect matching M' .

Define the graph $G'' = (L'', R'', E'')$ to be the subgraph of G induced by $L \setminus S, R \setminus N_G(S)$, namely

$$\begin{cases} L'' = L \setminus S \\ R'' = R \setminus N_G(S) \\ E'' = E \cap (L'' \times R'') \end{cases}$$

We show that G'' has a perfect matching.

Given any $S^* \subseteq L/S$, we have

$$|S| + |N_{G''}(S^*)| = |N_G(S)| + |N_{G''}(S^*)| = |N_G(S \cup S^*)| \geq |S^* \cup S| = |S| + |S^*|$$

showing that $|N_{G''}(S^*)| \geq |S^*|$ (this idea is based on the fact that if the claim is not true, then there are less neighbours of $S \cup S^*$ than the number of vertices in $S \cup S^*$).

Since, $|L \setminus S| = |R \setminus N_G(S)| < n + 1$, G'' has a perfect matching M'' by inductive hypothesis. Take $M = M' \cup M''$, this gives a perfect matching of G . $\square$

2.1.3. Algorithm to find a Perfect Matching.

We can now check whether a graph has a perfect matching by checking if satisfies Hall's Theorem. However, we will now provide an even more efficient algorithm to determine whether a graph has a perfect matching.

Definition 2.7. Given a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$, the adjacency matrix A of G is defined as a $n \times n$ matrix such that $A_{x,y} = 1$ if and only if $(x, y) \in E$, and $A_{x,y} = 0$ otherwise.

Algorithm 2.8. Given a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$.

- Compute the $n \times n$ adjacency matrix A
- Define $\tilde{A}$ to be an $n \times n$ matrix such that

$$(\tilde{A})_{ij} = \begin{cases} 0 & \text{if } (A)_{ij} = 0 \\ m \stackrel{R}{\leftarrow} [M] & \text{if } (A)_{ij} = 1 \end{cases}$$

- If $\det(\tilde{A}) = 0$, output no perfect matching. Else say yes perfect matching.

Before stating and proving the theorem, we will provide a lemma

Lemma 2.9. (Schwartz-Zippel Lemma) Let a subset of real numbers $A \subset \mathbb{R}$ satisfies that $|A| = M$, if $Q(y_1, \dots, y_k)$ is a non-zero polynomial of degree $\leq d$ then

$$\Pr_{\vec{y} \stackrel{R}{\leftarrow} A^k} [Q(\vec{y}) = 0] \leq \frac{d}{M}$$

Note: This is equivalent to saying $|\{\vec{y} \in A^k : Q(\vec{y}) = 0\}| \leq dM^{k-1}$

Proof. We prove by induction on k .

Base case: We have $k = 1$. If $Q(y_1)$ is a non-zero polynomial with degree $\leq d$ then there are at most d roots. Hence, A contains at most d roots, and

$$\Pr_{y_1 \in A} [Q(y_1) = 0] \leq \frac{d}{M}$$

Inductive step: Consider the above result works for polynomial Q with $k = n$ variables, for Q with $k = n + 1$ variables, we can write

$$Q(y_1, \dots, y_n, z) = Q_0(y_1, \dots, y_n) + Q_1(y_1, \dots, y_n)z + \dots + Q_t(y_1, \dots, y_n)z^t$$

where $Q_t(y_1, \dots, y_n)$ is a non-zero polynomial such that t is the largest. Moreover, for each $i \in [t]$, Q_i has a degree of at most $d - i$

We get

$$\begin{aligned} \Pr_{\vec{y} \leftarrow A^{n+1}} [Q(\vec{y}) = 0] &= \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q(\vec{y}, z) = 0] \\ &= \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q(\vec{y}, z) = 0 \cap Q_t(\vec{y}) = 0] + \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q(\vec{y}, z) = 0 \cap Q_t(\vec{y}) \neq 0] \\ &= \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q(\vec{y}, z) = 0 \mid Q_t(\vec{y}) = 0] \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q_t(\vec{y}) = 0] + \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q(\vec{y}, z) = 0 \mid Q_t(\vec{y}) \neq 0] \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q_t(\vec{y}) \neq 0] \\ &\leq \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q_t(\vec{y}) = 0] + \Pr_{\substack{\vec{y} \leftarrow A^n \\ z \leftarrow A}} [Q(\vec{y}, z) = 0 \mid Q_t(\vec{y}) \neq 0] \\ &= \Pr_{\vec{y} \leftarrow A^n} [Q_t(\vec{y}) = 0] + \Pr_{z \leftarrow A} [Q(\vec{y}, z) = 0 \mid Q_t(\vec{y}) \neq 0] \\ &\leq \frac{d-t}{M} + \Pr_{z \leftarrow A} [Q(\vec{y}, z) = 0 \mid Q_t(\vec{y}) \neq 0] \quad [Q_t \text{ have } d-t \text{ roots by IH}] \\ &\leq \frac{d-t}{M} + \frac{t}{M} \quad [Q \text{ have } t \text{ roots in } z] \\ &= \frac{d}{M} \end{aligned}$$

□

Theorem 2.10. Given a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$, *Algorithm 2.8* satisfies that

- If G has a perfect matching

$$\Pr[\text{Algorithm 2.8 outputs yes}] \geq 1 - \frac{n}{M}$$

- If G has no perfect matching

$$\Pr[\text{Algorithm 2.8 outputs no}] = 1$$

Proof. Given the adjacency matrix A and the matrix $\tilde{A}$, we have by definition

$$\det(\tilde{A}) = \sum_{\sigma: [n] \rightarrow [n]} (-1)^{\text{sign}(\sigma)} \prod_{i=1}^n \tilde{A}_{i, \sigma(i)}$$

- Suppose G has no perfect matching. Note that if there is any perfect matching in G , it can be written as some permutation $\sigma: [n] \rightarrow [n]$ from the left vertices to the right. In this case, for all

$i \in [n]$, $(i, \sigma(i)) \in E$. Therefore, since G has no perfect matching, for all permutation σ , there exists some i such that $(i, \sigma(i)) \notin E$. Therefore,

$$\det(\tilde{A}) = \sum_{\sigma: [n] \rightarrow [n]} (-1)^{\text{sign}(\sigma)} \prod_{i=1}^n \tilde{A}_{i, \sigma(i)} = \sum_{\sigma: [n] \rightarrow [n]} (-1)^{\text{sign}(\sigma)} \cdot 0 = 0$$

hence the algorithm would output no.

- Suppose G has a perfect matching, consider the polynomial $P(\{\tilde{A}_{ij} : A_{ij} = 1\})$ given by

$$P(\{\tilde{A}_{ij} : A_{ij} = 1\}) = \det(\tilde{A}) = \sum_{\sigma: [n] \rightarrow [n]} (-1)^{\text{sign}(\sigma)} \prod_{i=1}^n \tilde{A}_{i, \sigma(i)}$$

Note that P is a degree n polynomial where $\tilde{A}_{ij} \in [M]$ for all $A_{ij} = 1$. By Schwarz-Zippel Lemma, $\Pr[\det(\tilde{A}) = 0] \leq \frac{n}{M}$. Hence, the algorithm would output yes in probability more than $1 - \frac{n}{M}$. $\square$

2.1.4. A Matching-Based Magic Trick.

Question 2.11. There are two magicians M_1, M_2 and one volunteer.

1. The volunteer chooses 5 cards from a standard deck and gives them to M_2 .
2. M_1 reads out exactly 4 of these cards, in some order.
3. M_2 (who does not see the cards directly) must identify the missing fifth card.

How are M_1 and M_2 able to perform the trick, if no additional information is exchanged between the two magicians?

Before answering this question, let's consider an easier example to give some inspiration.

Example 2.12. Suppose instead that the volunteer picks only two cards but also tosses a coin, M_1 reads out both cards, and M_2 must guess which the side the coin landed on. It is possible for M_2 to encode heads or tails by choosing the order in which the two cards are read (read the card with the lower number first if heads and the other way if tails).

Remark 2.13. While we see the selected cards is represented as a set but it is actually an ordered tuple when M_1 reads out. Thus, by reading it in a specific way, this gives M_2 a way to decode and find out the fifth card.

To accomplish this, we can use a matching property between the set of all the possible ways of selecting five cards and the set of all possible ways M_1 reads out four cards so that M_2 may uniquely decode the fifth card.

We will use the following variant of Hall's Theorem

Theorem 2.14. (Hall's Theorem) Consider a bipartite graph $G = (L, R, E)$, suppose every $S \subseteq L$ satisfies

$$|N_G(S)| \geq |S|$$

then there is a matching of size $|L|$.

Proposition 2.15. Consider a set $[52]$ and a graph $G = (L, R, E)$ with

- $L = \binom{[52]}{5}$
- $R = \{(x_1, x_2, x_3, x_4) \in [52]^4 : x_i \text{ are distinct}\}$
- $E = \{(A, \vec{b} = (x_1, x_2, x_3, x_4) \in L \times R : \{x_1, x_2, x_3, x_4\} \subseteq A)\}$

There exists a matching that saturates all vertices of L . Equivalently, there exists an injection from $f: L \rightarrow R$ such that for all $f(x) = y, (x, y) \in E$.

Proof. Note that for any $A \in L$, we have $\deg(A) = \binom{5}{4} \cdot 4! = 120$. For any $\vec{b} \in R$, $\deg(\vec{b}) = 48$ (there are 48 remaining cards to choose from).

By Hall's Theorem, it is enough to show that for all $S \subseteq L$

$$|N(S)| \geq |S|$$

Given arbitrary S , we have that

$$e(S, N(S)) = |\{(a, b) \in E : a \in S \wedge b \in N(S)\}| = 120|S|$$

Similarly

$$e(S, N(S)) = |\{(a, b) \in E : a \in S \wedge b \in N(S)\}| \leq 48|N(S)|$$

Therefore, we get

$$120|S| \leq 48|N(S)| \implies |N(S)| \geq \frac{120}{48}|S| \geq |S|$$

Hence, there exists a matching that saturates all vertices of L , and there exists an injection from $f : L \rightarrow R$ such that for all $f(x) = y, (x, y) \in E$. $\square$

Remark 2.16. Therefore, given a deck of 52 cards, M_1, M_2 can do the following to perform the trick

- Label all cards from 1 – 52 so that the set of cards is denoted $[52]$.
- Fix one matching in advance.
- If M_1 receives five cards (which corresponds to some $A \in L$), M_1 reads the 4-card ordered tuple $\vec{b} = (x_1, x_2, x_3, x_4)$ matched to A .
- M_2 hears this tuple $\vec{b} \in R$, identifies the set A that match to $\vec{b}$, and outputs the missing card $A \setminus \{x_1, x_2, x_3, x_4\}$.

Because matching edges are disjoint on both sides, the decoding is unique.

Remark 2.17. This argument proves existence, but the matching could be very complicated and not human-friendly.

In practice, one would use explicitly memorise some rules. Moreover, the right degrees are much smaller than left degrees, so one can often impose extra constraints to reduce the right degrees and still keeping decodability.

2.2. Flows and Cuts in Graphs.

Definition 2.18. (Graph) $G = (V, E)$ where $E \subseteq V \times V$ is a **graph** if $(a, a) \notin E$ for all $a \in V$.

Definition 2.19. (Netflow of Vertices) Given a vertex $v \in V$,

$$\text{NetFlow}(v) = \sum_{(u,v) \in E} f(u, v) - \sum_{(v,u) \in E} f(v, u)$$

Given a subset $A \subseteq V$,

$$\text{NetFlow}(A) = \sum_{\substack{(u,v) \in E \\ u \notin A \\ v \in A}} f(u, v) - \sum_{\substack{(u,v) \in E \\ u \in A \\ v \notin A}} f(v, u)$$

Definition 2.20. Given a graph $G = (V, E)$, a **flow** in G is a function $f : E \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\forall v \in V, \text{NetFlow}(v) = 0$$

Definition 2.21. Given some distinct $s, t \in V$, an **$s - t$ flow** is a function $f : E \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\forall v \in V \setminus \{s, t\}, \text{NetFlow}(v) = 0$$

Moreover, there is no edge in the form of (u, s) or (t, u) for some $u \in V$.

In this case, f has a **flow value** of $\text{NetFlow}(t) = -\text{NetFlow}(s)$. We denote this as $\text{val}(f)$.

Remark 2.22. By the above definitions, we obtain

- For all graph $G = (V, E)$ and for all subset $A \subseteq V$

$$\sum_{\substack{(u,v) \in E \\ u \notin A \\ v \in A}} f(u, v) - \sum_{\substack{(v,u) \in E \\ u \notin A \\ v \in A}} f(v, u) = \sum_{v \in A} \left(\sum_{(u,v) \in E} f(u, v) - \sum_{(v,u) \in E} f(v, u) \right) \implies \text{NetFlow}(A) = \sum_{v \in A} \text{NetFlow}(v)$$

- Specifically to the $s - t$ graph

$$\text{NetFlow}(V) = \sum_{v \in V} \text{NetFlow}(v) = \text{NetFlow}(s) + \text{NetFlow}(t) = 0$$

Definition 2.23. (Capacity) Given a graph $G = (V, E)$, a **capacity function** on G is a function $C : E \rightarrow \mathbb{R}_{\geq 0}$.

Definition 2.24. (Maximum Flow) Given a graph $G = (V, E)$ with a capacity function $C : E \rightarrow \mathbb{R}_{\geq 0}$, the **max $s - t$ flow** is the flow $f : E \rightarrow \mathbb{R}_{\geq 0}$ such that $\forall e \in E, f(e) \leq C(e)$ and $\text{val}(f)$ is maximised.

In this case, we call $\text{val}(f)$ the **max $s - t$ flow value**.

Definition 2.25. Given a graph G , an **$s - t$ cut** is a subset $A \subseteq V$ such that $s \in A, t \notin A$.

In this case, together with a capacity function C , we denote

$$\text{CutValue}(A) = \sum_{\substack{(u,v) \in E \\ u \in A \\ v \notin A}} C(u, v)$$

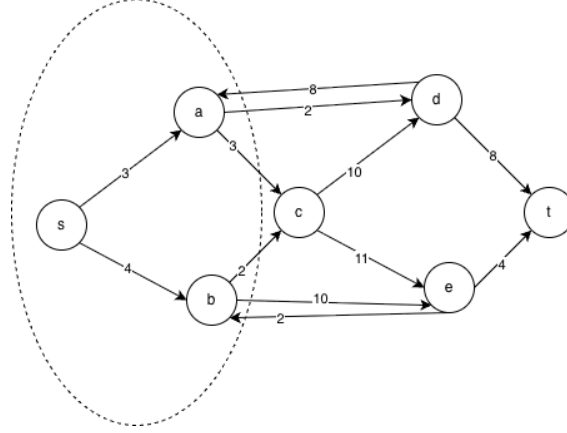
Remark 2.26. For every $s - t$ flow f obeying capacity constraints C , for any $s - t$ cut A , we have

$$\text{val}(f) \leq \text{CutValue}(A)$$

The idea of the above is simple, consider a cut A . Then we get

$$\text{val}(f) = -\text{NetFlow}(s) = -\text{NetFlow}(A) = \sum_{\substack{(v,u) \in E \\ u \notin A \\ v \in A}} f(v,u) - \sum_{\substack{(u,v) \in E \\ u \notin A \\ v \in A}} f(u,v) \leq \sum_{\substack{(v,u) \in E \\ u \notin A \\ v \in A}} f(v,u) \leq \sum_{\substack{(v,u) \in E \\ u \notin A \\ v \in A}} C(v,u) = \text{CutValue}(A)$$

This can be visualised by the following example



In this case, $A = \{s, a, b\}$, the maximum of $-\text{NetFlow}(A)$ can only satisfy when the arrows that flow out of A is maximised and no flow is flowing back to A . In this case, it is $2 + 3 + 2 + 10 = 17$.

2.2.1. Ford–Fulkerson algorithm.

It turns out that we know that if $\text{CutValue}(A)$ is minimised, $\text{val}(f) = \text{CutValue}(A)$. We will show this results by first providing the following algorithm.

Algorithm 2.27. Given a graph G with a capacity function C

1. Start with the zero flow f
2. Iterating the following
 - Create graph $G^* = (V, E^* = E_1 \cup E_2)$ where

$$E_1 = \{(u, v) \in V \times V : (u, v) \in E \wedge f(u, v) < C(u, v)\}$$

$$E_2 = \{(u, v) \in V \times V : (v, u) \in E \wedge f(v, u) > 0\}$$

where the first one tells us that there is capacity to increase the flow on the edge (u, v) . The second one tells us that there is capacity to decrease the flow on the edge (v, u) .

- Look for a path in G^* with the form $(s = v_0, v_1, \dots, v_k = t)$
- For some $\varepsilon > 0$, for each $i = 0, \dots, k - 1$

$$\begin{cases} \text{increase } f(v_i, v_{i+1}) \text{ by } \varepsilon & \text{if } (v_i, v_{i+1}) \in E_1 \\ \text{decrease } f(v_{i+1}, v_i) \text{ by } \varepsilon & \text{if } (v_{i+1}, v_i) \in E_1 \end{cases}$$

Remark 2.28. We will now assume C is a natural function, and ε always choose 1. The algorithm would obtain an $s - t$ flow with larger flow value by at least 1 until G^* has no $s - t$ path.

2.2.2. Max-Flow Min-Cut Theorem.

Theorem 2.29. (Max-Flow Min-Cut Theorem) *For all graph G with a capacity function C with a max $s - t$ flow f obeying the capacity constraint. Then*

$$\text{val}(f) = \min_A \text{CutValue}(A)$$

Proof. It is enough to find a cut A that satisfies the equality. Define $A = \{v : v \text{ can be reached from } s \text{ in } G^*\}$, we show that $\text{CutValue}(A) = \text{val}(f)$. We have

$$\text{val}(f) = -\text{NetFlow}(s) = -\text{NetFlow}(A) = \sum_{\substack{(u,v) \in E \\ u \in A \\ v \notin A}} f(u,v) - \sum_{\substack{(u,v) \in E \\ u \notin A \\ v \in A}} f(u,v) = \sum_{\substack{(u,v) \in E \\ u \in A, v \in A^c}} C(u,v) - 0 = \text{CutValue}(A)$$

There are no edges in G^* from A to A^c , otherwise, there is some vertices in A^c that is in A by definition. So $\forall u \in A, v \in A^c, f(u,v) = C(u,v)$, similarly, $\forall u \in A^c, v \in A, f(u,v) = 0$. $\square$

Corollary 2.30. *Given a graph G with **natural (rational) capacity function** C , there exists a **max-flow** f which is also a **natural (rational)** function, we can set the algorithm to increase flow by 1 each time (we can multiply the flow value to a common number to eliminate the denominator and gives us a natural capacity function). Consequently, the **max flow value** must be **natural (rational)**.*

There might also be a max-flow f which is not a natural function at the same time.

2.2.3. Application of Network Flow.

Definition 2.31. (Edge-disjointness) Given a graph $G = (V, E)$ with $s, t \in V$. Consider two paths from s to t , denoting $(s = u_0, u_1, \dots, u_n = t), (s = u_0, u_1, \dots, u_m = t)$ such that

$$\{(u_i, u_{i+1}) : i = 0, \dots, n-1\} \cap \{(v_i, v_{i+1}) : i = 0, \dots, m-1\} = \emptyset$$

we say these two paths are edge-disjoint.

Theorem 2.32. (Menger's Theorem) *Let G be an undirected graph with $s, t \in V$ then*

$\exists k$ edge-disjoint paths from s to $t \iff$ there is still a path from s to t if deleting any $k-1$ edges

Proof. $(\implies)$: Trivial.

$(\impliedby)$: Given $G = (V, E)$, together with $s, t \in V$, consider a directed graph $\tilde{G} = (\tilde{V}, \tilde{E})$ such that $\tilde{V} = V, \tilde{E} = E \cup \{(v, u) : (u, v) \in E\}$ (we make edges to go both ways). Take capacity function $C : \tilde{E} \rightarrow \mathbb{R}_{\geq 0}$ as 1 on all edges.

We show $\min_A \text{CutValue}(A) \geq k$. Indeed, if $\min_A \text{CutValue}(A) \leq k-1$ where A is the cut with the minimum cut value. There are $k-1$ edges (u, v) such that $u \in A, v \notin A$. Deleting all of these $k-1$ edges in G , there will be no edges from $s \in A$ to $t \notin A$. A contradiction.

Hence, G' has a max $s-t$ flow value of $\geq k$, which by **Corollary 2.30**, this flow value can be achieved by a natural number flow function $f : \tilde{E} \rightarrow \{0, 1\}$.

Now, we construct k edge-disjoint paths from the edges $\{(u, v) \in \tilde{E} : f(u, v) = 1\}$. To construct an edge-disjoint path by starting from s , we greedily search for an edge (u, v) that haven't searched before such that $f(u, v) = 1$ and $f(v, u) = 0$, repeat this process until we reach to t . Removing this path from the existing flow reduces the flow by 1. Repeating this process for k times we get k distinct edge-disjoint paths. $\square$

Theorem 2.33. *Given a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$, G has a perfect matching if and only if there exists a flow f with $\text{val}(f) = n$ in the directed graph $\tilde{G} = (\tilde{V}, \tilde{E})$ with $\tilde{V} = L \cup R \cup \{s, t\}, \tilde{E} = \{(s, u) : u \in L\} \cup E \cup \{(v, t) : v \in R\}$ with capacity function C to be 1 on all edges.*

Proof. $(\implies)$: We first show a matching M of size m corresponds to a flow f in $\tilde{G}$ with flow value of m . Given a matching M of size m , then define f to be

$$f(u, v) = \begin{cases} 1 & \text{if } u = s \vee v = t \vee (u, v) \in M \\ 0 & \text{otherwise} \end{cases}$$

This is a valid flow function that satisfies $\text{val}(f) = m$. Now, if there exists a perfect matching, there must be a flow f with $\text{val}(f) = n$. Moreover, there is no f such that $\text{val}(f) > n$ as taking the cut $A = \{s\}$ gives us $\text{CutValue}(A) = n$ gives us the maximum flow to be exactly n .

$(\impliedby)$: We now show a flow f in $\tilde{G}$ with flow value of m corresponds to a matching of size m . Given a flow f in $\tilde{G}$ with flow value of m , WLOG, we can assume $f : \tilde{E} \rightarrow \{0, 1\}$ by **Corollary 2.30**.

There exists m vertices $u_1, \dots, u_m \in L$ such that $f(s, u_i) = 1$ for all u_i . Hence, for each $u_i \in L$, there must be exactly one $v_j \in R$ such that $f(u_i, v_j) = 1$. Moreover, for each $v_j \in R$, there must be exactly one u_i such that $f(u_i, v_j) = 1$.

Hence, this gives us a matching of size m namely

$$M = \{(u_i, v_j) : f(u_i, v_j) = 1\}$$

Hence, if there is a flow f with $\text{val}(n)$, this gives us a matching of size n , which is a perfect matching. $\square$

Theorem 2.34. (Hall's Theorem revisited) *Consider a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$, if for all $S \subseteq L$, $|N_G(S)| \geq |S|$, where*

$$N_G(S) = \{v \in R : \exists u \in S \text{ s.t. } (u, v) \in E\}$$

then G has a perfect matching.

Proof. Given a bipartite graph $G = (L, R, E)$ with $|L| = |R| = n$, we show if G does not have a perfect matching, then we can find some $S \subseteq L$ that $|N_G(S)| < |S|$.

Suppose there is no perfect matching, we create $\tilde{G}$ in the same way as **Theorem 2.33**, except for all $(u, v) \in E$ where $u, v \notin \{s, t\}$, $C(u, v) = \infty$. The maximum flow of $\tilde{G}$ remains at most $n - 1$ (otherwise, we can find a perfect matching given that all other edges have capacity 1).

Hence, there exists a cut A such that $\text{CutValue}(A) \leq n - 1$. We can write $A = \{s\} \cup A_L \cup A_R$ with $A_L \subseteq L, A_R \subseteq R$. Since the cut value of A is finite, there is no $u \in A, v \notin A$ such that $(u, v) \in E$. Therefore, $N(A_L) \subseteq A \implies N(A_L) \subseteq A_R$.

We also have that

$$n - 1 \geq \text{CutValue}(A) = |L \setminus A_L| + |A_R| = n - |A_L| + |A_R| \implies |A_L| \geq |A_R| + 1$$

Hence, we have $|N(A_L)| < |A_L|$. $\square$

2.3. Ordered Sets.

Definition 2.35. (Partially/Totally Ordered Sets) A set S with a relation $\preceq$ is a **partially ordered set (poset)** if it satisfy

1. Reflexive: For all $a \in S, a \preceq a$
2. Transitive: For all $a, b, c \in S, a \preceq b, b \preceq c \implies a \preceq c$
3. Anti-symmetric: For all $a, b, a \preceq b \wedge b \preceq a \implies a = b$

$(S, \preceq)$ is a **totally ordered set** if the following additional property hold

1. For all $a, b, (a \preceq b) \vee (b \preceq a)$

Example 2.36. The following examples are posets

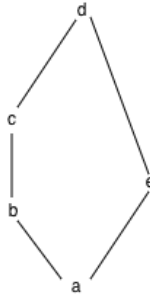
1. $(\mathbb{R}, \leq)$
2. $(\mathbb{R}^2, \text{coordinate-wise } \leq)$: when we say coordinate-wise, we have $(5, 7) \leq (6, 8), (5, 7) \not\leq (6, 6)$
3. $(\{a, b, \dots, z\}, \leq)$: the usual lexicographic order
4. (strings made out of $\{a, b, \dots, z\}, \leq$): the usual lexicographic order
5. $(\mathcal{P}(X), \subseteq)$: given some set X
6. $(\mathbb{N} \setminus \{0\}, |)$: the division relation

Definition 2.37. (Chain/Antichain) Given a poset $(S, \leq)$, a subset $A \subseteq S$ is a **chain** if for all $a, a' \in A, a \leq a' \vee a' \leq a$

We call A is an **anti-chain** if for all $a, a' \in A, (a \not\leq a') \wedge (a' \not\leq a)$

Remark 2.38. By definition, a singleton is a chain and an anti-chain.

Example 2.39. We will show how to draw a poset. We write the elements that can be compared into a line (to represent a chain), the lower the element, more frequent the element will be on the left hand side of the symbol $\preceq$



Definition 2.40. Given a poset $(S, \preceq)$, a **chain cover** of S is a set of chains $C_1, \dots, C_m$ such that $\bigcup_i C_i = S$.

In this case, the chain cover has a size of m .

2.3.1. Dilworth's Theorem.

We will now show a maximum size of anti-chain in a poset S is exactly the minimum number of chains that can cover S . To do this, we first easily see that the size of any anti-chain is bounded by the size of any chain cover.

Proposition 2.41. Given a poset $(S, \preceq)$, for all anti-chains $A \subseteq S$, for all chain cover $C_1, \dots, C_m$ such that $\bigcup_i C_i = S$ we have $|A| \leq m$.

Proof. Let $(S, \preceq)$ be a poset, $C_1, \dots, C_m$ be a chain cover. Let $A \subseteq S$ be an anti-chain. Then, each C_i contains at most one element in A . Hence, we have

$$|A| \leq m$$

□

Definition 2.42. Given a graph $G = (V, E)$, a subset $U \subseteq V$ is a vertex cover of G if for all $(u, v) \in E$

$$u \in U \vee v \in U$$

We also need to prove the following theorem

Theorem 2.43. (König's Theorem) *In a bipartite graph (L, R, E)*

$$\max\{\text{matching size}\} = \min\{\text{vertex cover size}\}$$

Proof. Suppose the maximum matching M has a size $|M|$. It is clear that there is no vertex cover U with $|U| < |M|$. Since M is a matching, each $v \in A$ can cover at most one edge of M . Hence, we will proceed by finding a vertex cover of size $|M|$.

The size of a maximum matching is the same as the maximum flow of the graph as defined in the same way as **Theorem 2.33** except for all $(u, v) \in E, C(u, v) = \infty$.

This is further equal to the minimum cut A of the graph. By writing $A = \{s\} \cup A_L \cup A_R$, certainly, there is no edges in the form (u, v) where $u \in A_L, v \in R \setminus A_R$. Hence

$$|M| = \text{CutValue}(A) = |L \setminus A_L| + |A_R|$$

where it is easy to see that $(L \setminus A_L) \sqcup A_R$ is a vertex cover of size $|M|$. For any edge (u, v) , we either have $u \in L \setminus A_L$, or we have $u \in A_L$ and $v \in A_R$. □

Theorem 2.44. (Dilworth's Theorem) *Given a partially ordered set $(S, \preceq)$, the*

$$\max\{\text{anti-chain size}\} = \min\{\text{number of chains cover } S\}$$

Proof. Suppose $|S| = n$, let S^-, S^+ be two copies of S , define a bipartite graph $G = (L, R, E)$ where

$$\begin{cases} L = S^- \\ R = S^+ \\ E = \{(x^-, y^+) : x \preceq y, x \neq y\} \end{cases}$$

By König's Theorem, there exists a maximum matching M and minimum vertex cover U , both in size m , i.e.

$$m = |M| = |U|$$

We show

1. There exists a chain cover of S with size $n - m$
2. There exist a anti-chain with size $\geq n - m$

(1) Consider the following algorithm

- Start with n chains $(\{a\})$ for each $a \in S$
- For each $(x^-, y^+) \in M$
 - Merge the chains of x and y

In this case, chain merge would repeat for m times hence giving us a chain cover of size $n - m$.

(2) Consider $U = U^- \cup U^+$ such that $U^- \subseteq S^-, U^+ \subseteq S^+$. Define

$$A = \{x \in S : x^- \notin U^- \wedge x^+ \notin U^+\}$$

we show the above set is an anti-chain.

Suppose $x \preceq y$ with $x \neq y$, we show we cannot have both $x \in A$ and $y \in A$. Note that $(x^-, y^+) \in E$. Hence, by the definition of a vertex cover we must have either have $x^- \in U^-$ or $y^+ \in U^+$.

If $x^- \in U^-$, then we have $x \notin A$. Similarly, if $y^+ \in U^+$, then $y \notin A$. Thus, in any case, at least one of them is not in A . Thus, A is an anti-chain.

Finally, we show that $|A| \geq n - m$. We have

$$|A| = |\{x \in S : x^- \notin U^- \wedge x^+ \notin U^+\}| \geq |S| - |U^-| - |U^+| = n - m$$

□

2.3.2. Sperner's Theorem and LYM Inequality.

Theorem 2.45. (Sperner's Theorem) *Given $X = [n]$ and a poset $(\mathcal{P}(X), \subseteq)$, the largest anti-chain of $(S, \preceq)$ has size at most $\binom{n}{\lfloor \frac{n}{2} \rfloor}$*

Proof. (By Probabilistic Method) Note that for any $\sigma \in S_n$ be a uniformly random permutation, consider arbitrary subset $B \subseteq [n]$ and define E_B be the event that $\{\sigma(1), \dots, \sigma(|B|)\} = B$. We have

$$\Pr[E_B] = \frac{|B|!}{n(n-1)\dots(n-|B|+1)} = \frac{1}{\binom{n}{|B|}}$$

Let $A \subseteq \mathcal{P}(X)$ be arbitrary anti-chain, it is clear that the events $\{E_B : B \in A\}$ are disjoint.

Therefore,

$$\sum_{B \in A} \Pr[E_B] = \sum_{B \in A} \frac{1}{\binom{n}{|B|}} \leq 1 \implies \sum_{B \in A} \frac{1}{\binom{n}{\lfloor \frac{n}{2} \rfloor}} \leq 1 \implies |A| \frac{1}{\binom{n}{\lfloor \frac{n}{2} \rfloor}} \leq 1 \implies |A| \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$$

□

Definition 2.46. Given a finite set S , we denote

$$\binom{S}{k} = \{S' \subseteq S : |S'| = k\}$$

if S is finite

$$\left| \binom{S}{k} \right| = \binom{|S|}{k}$$

Theorem 2.47. (LYM Inequality) *Given $X = \mathcal{P}([n])$, let A be an anti-chain in X . Let $A_k = A \cap \binom{[n]}{k}$, then*

$$\sum_{k=0}^n \frac{|A_k|}{\binom{n}{k}} \leq 1$$

Proof.

$$\begin{aligned} \sum_{k=0}^n \frac{|A_k|}{\binom{n}{k}} &= \sum_{B \in A} \frac{1}{\binom{n}{|B|}} \\ &\leq 1 \quad [\text{By Sperner's Theorem}] \end{aligned}$$

□

2.4. Necklaces.

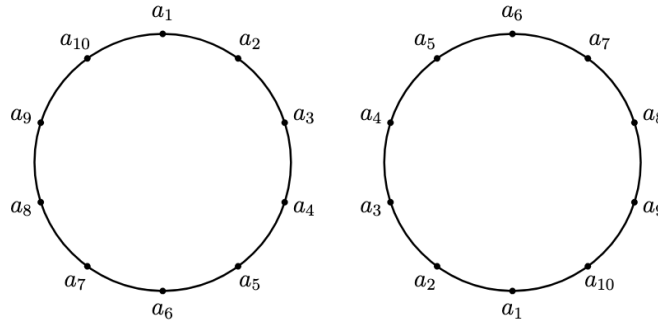
Definition 2.48. (Necklaces) Given a set Σ , a **necklace** of length n is a sequence of n elements from Σ . We denote this as

$$[x_1, \dots, x_n] : x_1, \dots, x_n \in \Sigma$$

Moreover, two sequences $[a_1, \dots, a_n], [b_1, \dots, b_n]$ are considered the same if there exists $i \in [n]$ such that

$$a_1 = b_i, a_2 = b_{i+1}, \dots, a_{n-i+1} = b_n, a_{n-i+2} = b_1, \dots, a_n = b_{i-1}$$

The following gives an example of two necklaces being equivalent. A sequence of necklace can be seen as writing the number in order on a circle (which exactly looks like a necklace!), where two necklaces are equivalent under rotation.



We won't study any combinatorial properties of necklaces. Instead, we will look at various results that we can prove using necklaces.

2.4.1. Application of Necklaces - Intersecting Families.

Definition 2.49. An intersecting family of sets is a family of sets $\mathcal{F}$ such that for any $A, B \in \mathcal{F}$,

$$A \cap B \neq \emptyset$$

Question 2.50. If $\mathcal{F} \subseteq \mathcal{P}([n])$, and $\mathcal{F}$ is an intersecting family, how big can $\mathcal{F}$ be?

Here are some examples of $\mathcal{F} \subseteq \mathcal{P}([n])$ which is also an intersecting family.

Example 2.51.

1. Consider

$$\mathcal{F}_1 = \{A \cup \{n\}, A \in \mathcal{P}([n-1])\}$$

In this case

$$|\mathcal{F}_1| = 2^{n-1}$$

2. Consider arbitrary c that satisfies $c \geq \frac{n}{2} + 1$

$$\mathcal{F}_2 = \left\{ \begin{matrix} [n] \\ c \end{matrix} \right\}$$

It turns out that, we get

$$|\mathcal{F}_2| < 2^{n-1}$$

3. Consider

$$\mathcal{F}_3 = \{A \in \mathcal{P}([n]) : \{1, 2\} \subseteq A \vee \{1, 3\} \subseteq A \vee \{2, 3\} \subseteq A\}$$

We have

$$|\mathcal{F}_3| = 4 \times 2^{n-3} = 2^{n-1}$$

We have shown in some of the examples above the size can reach 2^{n-1} . It turns out this is the maximum we can get.

Theorem 2.52. *If $\mathcal{F} \subseteq \mathcal{P}([n])$, and $\mathcal{F}$ is an intersecting family. Then $|\mathcal{F}| \leq 2^{n-1}$.*

Proof. Partition $\mathcal{P}([n])$ into 2^{n-1} pairs of the form $\{A, A^c\}$.

Let each pair to be the pigeonholes. Let the sets in $\mathcal{F}$ to be the pigeons. For every set in $\mathcal{F}$, we send to the corresponding pair that has it.

If $|\mathcal{F}| \geq 2^{n-1} + 1$, then there exists some $A \in \mathcal{P}([n])$ such that $A, A^c \in \mathcal{F}$, a contradiction. $\square$

Question 2.53. If $\mathcal{F} \subseteq \binom{[n]}{k}$, and $\mathcal{F}$ is an intersecting family, how big can $\mathcal{F}$ be?

Here are some examples of $\mathcal{F} \subseteq \binom{[n]}{k}$ which is also an intersecting family.

Example 2.54.

1. Consider

$$\mathcal{F}_1 = \left\{ A \in \binom{[n]}{k} : 1 \in A \right\}$$

In this case

$$|\mathcal{F}_1| = \binom{n-1}{k-1}$$

2. Trivial case: consider $k > \frac{n}{2}$, we can take the largest possible subset

$$\mathcal{F}_2 = \binom{[n]}{k}$$

and in this case

$$|\mathcal{F}| = \binom{n}{k}$$

3. Consider

$$\mathcal{F}_3 = \left\{ A \in \binom{[n]}{k} : A \cap \{1, 2, 3\} \in \{\{1, 2\}, \{1, 3\}, \{2, 3\}\} \right\}$$

In this case

$$|\mathcal{F}_3| = 3 \times \binom{n-3}{k-2}$$

It also turns out that

$$\binom{n-3}{k-2} \in \Theta(n^{k-2}) \leq \Theta(n^{k-1}) \ni \binom{n-1}{k-1}$$

Hence, the first example has the largest intersecting family.

Again, we have shown in some of the examples above the size can reach $\binom{n-1}{k-1}$. It turns out this is the maximum we can get.

Theorem 2.55. (Erdős-Ko-Rado) *If $\mathcal{F} \subseteq \binom{[n]}{k}$ such that $k < \frac{n}{2}$ and $\mathcal{F}$ is an intersecting family. Then $|\mathcal{F}| \leq \binom{n-1}{k-1}$.*

Proof. (By Probabilistic Method) For any intersecting family $\mathcal{F} \subseteq \binom{[n]}{k}$.

Pick a necklace $[a_1, \dots, a_n]$ with $\{a_1, \dots, a_n\} = \{1, \dots, n\}$ uniformly at random. This can be seen as picking $\sigma \in S_n$ uniform at random and consider $N = [\sigma(1), \dots, \sigma(n)]$.

Define

$\mathcal{F}_N = \{A \in \mathcal{F} : \text{there exists a contiguous intervals of } N = [\sigma(1), \dots, \sigma(n)] \text{ whose set of symbols is exactly } A\}$

For each $A \in \mathcal{F}$, denote the indicator random variable X_A

$$X_A = \begin{cases} 1 & \text{if } A \in \mathcal{F}_N \\ 0 & \text{if } A \notin \mathcal{F}_N \end{cases}$$

Then,

$$E[X_A] = \Pr_N[A \in \mathcal{F}_N] = \frac{n}{\binom{n}{k}}$$

Define the random variable $X = |\mathcal{F}_N|$. Therefore

$$X = \sum_{A \in \mathcal{F}} X_A \implies E[X] = \sum_{A \in \mathcal{F}} E[X_A] = \sum_{A \in \mathcal{F}} \frac{n}{\binom{n}{k}} = |\mathcal{F}| \frac{n}{\binom{n}{k}}$$

Now, we also show $X \leq k$ always. Note that $\mathcal{F}_N$ is an intersecting family of k -intervals on the necklace. If we have $k+1$ distinct sets in $\mathcal{F}_N$ that are intervals on the necklace then since $k < \frac{n}{2}$, some two sets are disjoint. A contradiction. So at most k pairwise intersecting sets in $\mathcal{F}$ can be contiguous sequences of the necklace.

Therefore, we have

$$|\mathcal{F}| \frac{n}{\binom{n}{k}} = E[X] \leq k \implies |\mathcal{F}| \leq \binom{n}{k} \frac{k}{n} = \binom{n-1}{k-1}$$

□

Remark 2.56. The intuition of this proof can be seen as averaging over overlapping necklaces. Each necklace containing at most k sets of $\mathcal{F}$, and every set is covered equally often.

2.4.2. Application of Necklaces - Fermat's Little Theorem.

Recall in group theory. We have the following definitions

Definition 2.57. Let G be a group acting on a set S . For $s \in S$, we define the **orbit** of s is

$$\text{orb}(s) = \{gs : g \in G\} \subseteq S$$

The **stabiliser** of s is

$$\text{stab}(s) = \{g \in G : gs = s\} \subseteq G$$

Note that $\text{stab}(s)$ is a subgroup of G .

Theorem 2.58. (Orbit-Stabiliser) *For any $s \in S$*

$$|\text{orb}(s)| = \frac{|G|}{|\text{stab}(s)|}$$

Definition 2.59. For a string $x = (x_1, \dots, x_n)$, define $R^i x$ as rotation of the string by i .

Thus, if $0 \leq i \leq n$, $R^i(x) = (x_{i+1}, \dots, x_n, x_1, \dots, x_i)$.

Theorem 2.60. (Fermat's Little Theorem) *For any prime number p , then for any integer a ,*

$$a^p \equiv a \pmod{p}$$

If $p \nmid a$ then we have

$$a^{p-1} \equiv 1 \pmod{p}$$

We should know the classical proof using the multiplicative group $\mathbb{Z}_p^* = \{1, \dots, p-1\}$ already when we study group theory. Instead, we will prove it using necklaces.

Proof. (Using Necklaces)

Taking arbitrary positive integer a , any prime n . Consider $\Sigma = \{0, 1, \dots, a-1\}$, there are a^p of length p Σ -strings in total. The set of strings can be partitioned based on which necklace each string belongs to. We will look at how many necklaces are there with length n .

Consider the group $\mathbb{Z}_p$ acts on Σ^p defined by the rotation action as **Definition 2.59**. For any string $x \in \Sigma^p$, define

$$\text{Periods}(x) = \{i \in \mathbb{Z}_p : R^i(x) = x\}$$

where $\mathbb{Z}_p = \{0, 1, \dots, p-1\}$ is a group under modular addition. Certainly, $\text{Periods}(x)$ is a subgroup of $\mathbb{Z}_p$.

If we consider the orbit $\text{orb}(x)$ and the stabiliser $\text{stab}(x)$ of x , it is easy to check that $\text{stab}(x) = \text{Periods}(x)$, by Orbit-Stabiliser Theorem

$$|\text{orb}(x)| = \frac{|\mathbb{Z}_p|}{|\text{stab}(x)|} = \frac{p}{|\text{Periods}(x)|}$$

Since p is prime, we get

$$|\text{Periods}(x)| = \begin{cases} p & \text{if } x \in \{(0, \dots, 0), (1, \dots, 1), \dots, (a-1, \dots, a-1)\} \\ 1 & \text{otherwise} \end{cases}$$

Therefore,

$$|\text{orb}(x)| = \begin{cases} 1 & \text{if } x \in \{(0, \dots, 0), (1, \dots, 1), \dots, (a-1, \dots, a-1)\} \\ n & \text{otherwise} \end{cases}$$

Therefore, if we partition Σ^p (with size a^p) into distinct orbits, we have

- a orbits of size 1
- $\frac{a^p - a}{p}$ orbits of size p (for some constant c)

Since each orbit corresponds to a distinct necklace, we must have

$$\frac{a^p - a}{p} \in \mathbb{Z} \implies a^p - a \equiv 0 \pmod{p} \implies a^p \equiv a \pmod{p}$$

For the case a is negative, we can use the result when a is positive and it is a simple algebraic manipulation. Except for the case where $p = 2$ or $a = 0$, we can prove it using simple algebra, without using necklaces. $\square$

Remark 2.61. There is another method of proving Fermat's Little Theorem, using induction on a . Suppose that $a^n \equiv a \pmod{n}$ and we want to show that $(a+1)^n \equiv (a+1) \pmod{n}$. We can expand $(a+1)^n$ using the Binomial Theorem. Except for the terms 1 and a^n , the coefficient of every other term is divisible by n . Hence, $(a+1)^n \equiv a^n + 1 \equiv a + 1 \pmod{n}$ by the inductive hypothesis.

2.4.3. Counting Necklaces.

We have proved Fermat's Little Theorem, which gives a way of counting necklaces. However, this assume that n is prime. We will give a more generalised result than the Fermat's Little Theorem where in this case, n doesn't have to be prime. This requires the following lemma

Lemma 2.62. (Burnside's Lemma) *Given arbitrary finite group G acting on S , the number of orbits of G is the average number of fixed points. That is*

$$\# \text{ of orbits} = \frac{1}{|G|} \cdot \sum_{g \in G} \text{Fix}(g)$$

where

$$\text{Fix}(g) = |\{s \in S : gs = s\}|$$

Proof. We have

$$\begin{aligned}
\sum_{g \in G} \text{Fix}(g) &= \sum_{g \in G} \sum_{s \in S} \mathbb{1}_{gs=s} \\
&= \sum_{s \in S} \left(\sum_{g \in G} \mathbb{1}_{gs=s} \right) \\
&= \sum_{s \in S} |\text{stab}(s)| \quad [\text{By definition of a stabliser}] \\
&= \sum_{s \in S} \frac{|G|}{|\text{orb}(s)|} \quad [\text{By Orbit-Stabliser}] \\
&= |G| \sum_{s \in S} \frac{1}{|\text{orb}(s)|} \\
&= |G| \sum_{\text{orbit } O} \left(\sum_{s \in O} \frac{1}{|O|} \right) \\
&= |G| \sum_{\text{orbit } O} 1 \\
&= |G| \cdot \# \text{ of orbits}
\end{aligned}$$

□

Now we will provide the theorem.

Theorem 2.63. *Consider arbitrary positive integers n and a , then*

$$n \mid \sum_{d \mid n} a^d \varphi\left(\frac{n}{d}\right)$$

where φ is the Euler's Tortient Function

Proof. Consider arbitrary a, n , denote $\Sigma = \{0, 1, \dots, a-1\}$. We let the group $\mathbb{Z}_n$ acts on Σ^n by the rotation action. That is

$$i \cdot (x_1, \dots, x_n) = (x_{i+1}, \dots, x_n, x_1, \dots, x_i)$$

Therefore, we get

$$\begin{aligned}
|\text{Fix}(i)| &= |\{x \in \Sigma^n : i \cdot x = x\}| \\
&= \# \text{ strings that are } i\text{-periodic} \\
&= \# \text{ strings that are } \gcd(i, n)\text{-periodic} \quad [\text{By Bezout's Identity}] \\
&= a^{\gcd(i, n)}
\end{aligned}$$

Therefore,

$$\begin{aligned}
 \# \text{ of orbits} &= \frac{1}{|Z_n|} \sum_{i \in Z_n} \text{Fix}(i) \\
 &= \frac{1}{n} \sum_{i=0}^{n-1} a^{\gcd(i,n)} \\
 &= \frac{1}{n} \sum_{d|n} a^d (\# \text{ of } i\text{'s such that } \gcd(i,n) = d) \\
 &= \frac{1}{n} \sum_{d|n} a^d \left(\# \text{ of } i\text{'s such that } \gcd\left(\frac{i}{d}, \frac{n}{d}\right) = 1 \right) \\
 &= \frac{1}{n} \sum_{d|n} a^d \varphi\left(\frac{n}{d}\right)
 \end{aligned}$$

Since the number of orbits must be an integer, we have

$$n \mid \sum_{d|n} a^d \varphi\left(\frac{n}{d}\right)$$

□

Example 2.64. If we consider $n = pq$ where p, q are prime then we get

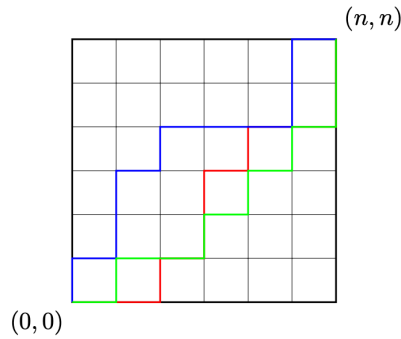
$$pq \mid a\varphi(pq) + a^p\varphi(q) + a^q\varphi(p) + a^{pq}\varphi(1) \implies pq \mid a(p-1)(q-1) + a^p(q-1) + a^q(p-1) + a^{pq}$$

2.4.4. Application of Necklaces - Lattice Paths.

Definition 2.65. A **lattice path** of an $n \times n$ **lattice** is a path that starts at $(0,0)$ and ends at (n,n) . Moreover, for each step, we can only do the following

- Move up $(x, y) \rightarrow (x, y+1)$
- Move right $(x, y) \rightarrow (x+1, y)$

Here is an example of three lattice paths.



Proposition 2.66. The total number of lattice paths to (n, n) is

$$\binom{2n}{n}$$

Recall the Sterlings Approximation

$$n! = \Theta\left(\sqrt{n} \cdot \left(\frac{n}{e}\right)^n\right)$$

We get

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \Theta\left(\frac{\sqrt{2n} \cdot \left(\frac{2n}{e}\right)^{2n}}{\left(\sqrt{n} \cdot \left(\frac{n}{e}\right)^n\right)^2}\right) = \Theta\left(\frac{1}{\sqrt{n}} 2^{2n}\right)$$

We can interpret this quantity in terms of coin tosses. Suppose X is the random variable corresponding to the number of heads when tossing $2n$ fair coins. Then,

$$E[X] = n, \text{Var}[X] = \frac{n}{2}$$

so the standard deviation is $\Theta(\sqrt{n})$.

There are 2^{2n} outcomes of tossing $2n$ coins. And there are $\binom{2n}{n}$ outcomes where we get exactly n heads. Therefore, it follows that

$$\binom{2n}{n} = 2^{2n} \Pr(X = n) = \Theta\left(\frac{1}{\sqrt{n}} 2^{2n}\right)$$

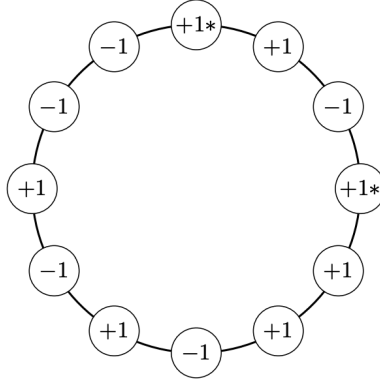
Definition 2.67. The n -th Catalan number C_n is the number of lattice paths from $(0,0)$ to (n,n) that never go above the diagonal (i.e. we always have $x \leq y$)

It seems very unlikely that this would occur, and it is easy to shoot up the y -value, so one might expect the Catalan number to be exponentially smaller than $\binom{2n}{n}$. But it turns out, we will see that it turns polynomially smaller than $\binom{2n}{n}$. To do this, we need the following result

Definition 2.68. Consider a necklace consisting of beads labeled as $+1$ or -1 .

We call a bead is **special** if all clockwise partial sums starting from that bead are **positive**.

Here is an example of a necklace where special beads are marked with $*$.



Remark 2.69. We use clockwise in the definition of **special beads** for simplicity.

We can also do anti-clockwise, as long as it is consistent.

Theorem 2.70. (Ballot Theorem) Suppose a, b are the numbers of $+1, -1$ beads on a necklace respectively. Then

$$\# \text{ of special beads} = \max(a - b, 0)$$

Proof. This is an induction proof.

Observe that if there are two consecutive $+1, -1$ beads (in either order), deleting them does not affect the specialness of other beads. To explain this a bit more requires a bit of trivial case by case analysis.

Repeating this process, the necklace eventually reduces to a collection of beads of a single type. If only $+1$ beads remain, there are exactly $a - b$ beads of $+1$, and they are all special. If not, no bead is special. Hence, the number of special beads is $\max(a - b, 0)$. $\square$

Theorem 2.71. (Catalan Number)

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

Proof. We can see a $+1$ bead corresponds to moving one step right and a -1 bead to moving one step up on an $n \times n$ lattice.

Therefore, we can see the lattice paths from $(0, 0)$ to (n, n) that never go above the diagonal as the strings made out of n $(+1)$'s and n (-1) 's such that the partial sums are always non-negative. We will find this number.

We will first find the number of strings made out of $n+1$ $(+1)$'s and n (-1) 's such that the partial sums are always positive.

Consider necklaces with $n+1$ $(+1)$'s and n (-1) 's. Each necklace has $2n+1$ linear representations as a string in $\{+1, -1\}^{2n+1}$ (because of rotations). We get

$$\# \text{ of representations} = \binom{2n+1}{n+1}, \# \text{ of necklaces} = \frac{1}{2n+1} \binom{2n+1}{n+1} = \frac{1}{n+1} \binom{2n}{n}$$

Moreover, by Ballot Theorem, there is exactly one special bead. Consider the string representation of this necklace where this special bead is placed at the front. By definition, the partial sums that starts at this special beads are always positive. Since for all necklaces, there is exactly one string representation such that all partial sums are always positive, and the string representations are distinct with different necklaces, there are exactly

$$\frac{1}{n+1} \binom{2n}{n}$$

such string made out of $n+1$ $(+1)$'s and n (-1) 's such that the partial sums are always positive.

There is a one-to-one correspondance between the strings made out of n $(+1)$'s and n (-1) 's such that the partial sums are always non-negative and the number of strings made out of $n+1$ $(+1)$'s and n (-1) 's such that the partial sums are always positive. We simply add a $+1$ to the front of each string in the former to obtain a string in the latter and vice versa. Hence, there are

$$\frac{1}{n+1} \binom{2n}{n}$$

lattices paths from $(0, 0)$ to (n, n) that never go above the diagonal. □

3. PROBABLISTIC METHODS

3.1. Basic Tools.

3.1.1. Probability.

Theorem 3.1. (Markov's Inequality) *Let X be a non-negative random variable. Then, for all $t > 0$*

$$\Pr[X \geq t] \leq \frac{E[X]}{t}$$

Proof. We have

$$E[X] = \sum_a a \Pr[X = a] = \sum_{a \geq t} a \Pr[X = a] + \sum_{0 \leq a < t} a \Pr[X = a] \leq \sum_{a \geq t} t \Pr[X = a] + 0 = t \Pr[X \geq t]$$

Proof for continuous random variables are similar. $\square$

Theorem 3.2. (Chebyshev's Inequality) *Let X be a random variable and $E[X] = \mu$, $\text{Var}(X) = E[(X - \mu)^2]$. Then, for all $t > 0$*

$$\Pr[|X - \mu| \geq t] \leq \frac{\text{Var}(X)}{t^2}$$

Remark 3.3. Chebyshev's Inequality is considered to be a better bound as it considers the second moment instead of the first.

Proof. We simply apply Markov's Inequality on the non-negative random variable $(X - \mu)^2$

$$\Pr[|X - \mu| \geq t] = \Pr[(X - \mu)^2 \geq t^2] \leq \frac{E[(X - \mu)^2]}{t^2} = \frac{\text{Var}(X)}{t^2}$$

$\square$

Remark 3.4. For an even better bound than the Chebyshev's Inequality, we can consider even higher moments. For example, we can consider the fourth moment by replacing $\text{Var}(X) = E[(X - \mu)^2]$ with $\mu_4(X) = E[(X - \mu)^4]$. And it is very similar.

Theorem 3.5. (Fourth-Moment Inequality) *Let X be a random variable. Let $E[X] = \mu$. Then, for any $t > 0$*

$$\Pr[|X - \mu| \geq t] \leq \frac{\mu_4[X]}{t^4}$$

Proof. Proof is similar as Chebeshev's Inequality.

We simply apply Markov's Inequality on the non-negative random variable $(X - \mu)^4$

$$\Pr[|X - \mu| \geq t] = \Pr[(X - \mu)^4 \geq t^4] \leq \frac{E[(X - \mu)^4]}{t^4} = \frac{\mu_4[X]}{t^4}$$

$\square$

Example 3.6. Let $X_1, \dots, X_n$ be independent coins such that for all X_i

$$\begin{cases} \Pr[X_i = 1] = 0.1 \\ \Pr[X_i = 0] = 0.9 \end{cases}$$

Let $X = \sum_i X_i$, we have

$$E[X] = \sum E[X_i] = 0.1n$$

Hence, by Markov's Inequality

$$\Pr[X \geq 0.2n] \leq \frac{E[X]}{0.2n} = \frac{0.1n}{0.2n} = \frac{1}{2}$$

Moreover,

$$\begin{aligned}
\text{Var}[X] &= E[(X - \mu)^2] \\
&= E\left[\left(\sum_{i=1}^n X_i - 0.1n\right)^2\right] \\
&= E\left[\left(\sum_{i=1}^n (X_i - 0.1)\right)^2\right] \\
&= E\left[\sum_{i=1}^n \sum_{j=1}^n (X_i - 0.1)(X_j - 0.1)\right] \\
&= \sum_{i=1}^n \sum_{j=1}^n E[(X_i - 0.1)(X_j - 0.1)] \\
&= \sum_{i=1}^n E[(X_i - 0.1)^2] + \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} E[X_i - 0.1]E[X_j - 0.1] \quad [\text{By Independence}] \\
&= \sum_{i=1}^n E[(X_i - 0.1)^2] \\
&\leq \sum_{i=1}^n 0.9^2 \quad [-0.9 \leq X_i - 0.1 \leq 0.9] \\
&\leq \sum_{i=1}^n 1 \\
&= n
\end{aligned}$$

By Chebyshev's Inequality

$$\Pr[|X - 0.1n| \geq t] \leq \frac{n}{t^2}$$

Setting $t = 0.1n$, we get

$$\Pr[|X - 0.1n| \geq 0.1n] \leq \frac{n}{(0.1n)^2} \implies \Pr[X \geq 0.2n] \leq \frac{100}{n} \in \mathcal{O}\left(\frac{1}{n}\right)$$

Finally, we have Moreover,

$$\begin{aligned}
\mu_4[X] &= E[(X - \mu)^4] \\
&= E\left[\left(\sum_{i=1}^n X_i - 0.1n\right)^4\right] \\
&= E\left[\left(\sum_{i=1}^n (X_i - 0.1)\right)^4\right] \\
&= E\left[\sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n (X_i - 0.1)(X_j - 0.1)(X_k - 0.1)(X_l - 0.1)\right]
\end{aligned}$$

Any term containing a linear term of $(X_i - 0.1)$ will be evaluated to 0. For example $E[(X_7 - 0.1)(X_3 - 0.1)^3] = E[(X_7 - 0.1)]E[(X_3 - 0.1)^3] = 0E[(X_3 - 0.1)^3] = 0$ by independence. Therefore, we get

$$\begin{aligned}
\mu_4[X] &= \sum_{i=1}^n E[(X_i - 0.1)^4] + 6 \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} E[(X_i - 0.1)^2 (X_j - 0.1)^2] \\
&\leq \sum_{i=1}^n E[(X_i - 0.1)^4] + 6 \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} E[(X_i - 0.1)^2] E[(X_j - 0.1)^2] \quad [\text{By Independence}] \\
&\leq \sum_{i=1}^n 0.9^4 + 6 \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} 0.9^4 [-0.9 \leq X_i - 0.1 \leq 0.9] \\
&\leq \sum_{i=1}^n 1 + 6 \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} 1 \\
&\leq n + 6n^2
\end{aligned}$$

By the fourth-moment inequality

$$\Pr[|X - \mu| \geq t] \leq \frac{n + 6n^2}{t^4}$$

Setting $t = 0.1n$, we have

$$\Pr[|X - 0.1n| \geq 0.1n] \leq \frac{n + 6n^2}{(0.1n)^4} \implies \Pr[X \geq 0.2n] \leq 10^4 \frac{1}{n^2} \left(\frac{1}{n} + 6 \right) \in \mathcal{O}\left(\frac{1}{n^2}\right)$$

Definition 3.7. (Independence) A collection of random variables $X_1, \dots, X_k$ are independent if for all $x_1, \dots, x_k$

$$\Pr[X_1 = x_1 \wedge \dots \wedge X_k = x_k] = \prod_{i=1}^k \Pr[X_i = x_i]$$

Proposition 3.8. (Expectation of Independent Random Variables) Let $X_1, \dots, X_k$ be independent random variables so that $E[X_i]$ is finite for all X_i . Then

$$E\left[\prod_{i=1}^k X_i\right] = \prod_{i=1}^k E[X_i]$$

Proof. We have

$$\begin{aligned}
E\left[\prod_{i=1}^k X_i\right] &= \sum_b \Pr\left[\left[\prod_{i=1}^k X_i\right] = b\right] \cdot b \\
&= \sum_b \left(\sum_{\substack{a_1, \dots, a_k \\ \prod_{i=1}^k a_i = b}} \Pr[X_1 = a_1 \wedge \dots \wedge X_k = a_k] \right) \cdot b \\
&= \sum_b \left(\sum_{\substack{a_1, \dots, a_k \\ \prod_{i=1}^k a_i = b}} \Pr[X_1 = a_1 \wedge \dots \wedge X_k = a_k] \cdot \prod_{i=1}^k a_i \right) \\
&= \sum_{a_1, \dots, a_k} \left(\prod_{i=1}^k \Pr[X_i = a_i] \cdot \prod_{i=1}^k a_i \right) \\
&= \sum_{a_1, \dots, a_k} \left(\prod_{i=1}^k (\Pr[X_i = a_i] \cdot a_i) \right) \\
&= \prod_{i=1}^k \left(\sum_{a_i} (\Pr[X_i = a_i] \cdot a_i) \right) \\
&= \prod_{i=1}^k E[X_i]
\end{aligned}$$

□

Definition 3.9. Let $X_1, \dots, X_n$ be random variables. For some $k \leq n$, $X_1, \dots, X_n$ are said to be k -wise independent if any k of the X_i 's are independent.

Remark 3.10. Independence versus pairwise independence is analogous to vectors being linearly independent versus vectors being pairwise linearly independent. For example, the vectors $\vec{v} = (0, 1)$, $\vec{w} = (1, 0)$, $\vec{x} = (2, 1)$ are not linearly independent but are pairwise linearly independent.

Remark 3.11. Chebyshev's Inequality requires only pairwise independence.

3.1.2. Linear Algebra over Finite Fields.

Definition 3.12. We denote $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ to be a field.

Note: Since this is a field, we must have p is prime.

Proposition 3.13. Consider vectors $v_1, \dots, v_n \in \mathbb{F}_p^m$ where $n \leq m$, pick $Y \in \mathbb{F}_p^m$ uniformly at random.

Define $\mathbb{F}_p$ -random variables $X_1, \dots, X_n$ to satisfy

$$X_i = \langle Y, v_i \rangle = \sum_{j=1}^m y_j \cdot (v_i)_j$$

If $v_1, \dots, v_n$ are linearly independent, then $X_1, \dots, X_n$ are independent random variables.

Proof. Let $V \in \mathbb{F}_p^{n \times m}$ be the matrix with rows $v_1, \dots, v_n$

$$V = \begin{pmatrix} - & v_1 & - \\ - & v_2 & - \\ \vdots & \vdots & \vdots \\ - & v_n & - \end{pmatrix}$$

Computing the joint probability

$$\begin{aligned}
\Pr[X_1 = a_1, \dots, x_n = a_n] &= \Pr[\langle Y, v_1 \rangle = a_1, \dots, \langle Y, v_n \rangle = a_n] \\
&= \frac{\left| \left\{ \mathbf{x} \in \mathbb{F}_p^m : \begin{bmatrix} V\mathbf{x} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \end{bmatrix} \right\} \right|}{p^m}
\end{aligned}$$

We now need to find the numerator.

Since $v_1, \dots, v_n$ are linearly independent, $\text{img}(V) = \mathbb{F}_p^n$, there exists a vector $w \in \mathbb{F}_p^m$ such that

$$Vw = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

For all $b \in \mathbb{F}_p^n$, define the set $S_b = \{y : Vy = b\}$. We know that S_0 is a linear subspace, by Rank-Nullity,

$$\dim(S_0) = m - \text{rank}(V) = m - n$$

Finally, we have

$$S_a = \{w + s : s \in S_0\} \implies |S_a| = |S_0| = p^{m-n}$$

Therefore, we get

$$\Pr[X_1 = a_1, \dots, x_n = a_n] = \frac{p^{m-n}}{p^m} = \frac{1}{p^n}$$

Moreover, for all each X_i , the distribution is uniform, we have

$$\Pr[X_i = a_i] = \frac{1}{p}$$

Therefore,

$$\Pr[X_1 = a_1, \dots, X_n = a_n] = \frac{1}{p^n} = \left(\frac{1}{p}\right)^n = \prod_{i=1}^n \Pr[X_i = a_i]$$

showing independence. □

Remark 3.14. If this procedure were carried out over the real vectors, orthogonality of the vectors $v_1, \dots, v_n$ would be necessary to get the same result.

Definition 3.15. A collection of vectors $v_1, \dots, v_n \in V$ is k -wise linearly independent if $k \leq n$ and any k vectors are linearly independent.

Corollary 3.16. If $v_1, \dots, v_n$ are k -wise linearly independent, then $X_1, \dots, X_n$ are k -wise independent random variables.

3.1.3. Generating Independent Random Variables over Finite Fields.

Inspired by the previous subsection, we will generate random variables that are independent where each random variable corresponds to a vector v_i . We will focus on pairwise independence in this case.

Remark 3.17. The largest number of random variables we can produce is related to the largest number of pairwise linearly independent vectors we can find.

Let $v_1, \dots, v_{2^m-1}$ denote all the nonzero possible vectors in $\mathbb{F}_2^m$. Note that in $\mathbb{F}_2^m$, any two distinct vector $v_i, v_j \in \mathbb{F}_2^m$ are linearly independent.

Let $Y \in \mathbb{F}_2^m$ be chosen uniformly at random. Define $(X_1, \dots, X_{2^m-1}) = (\langle Y, v_1 \rangle, \dots, \langle Y, v_{2^m-1} \rangle)$.

Hence, by Corollary 3.16, $X_1, \dots, X_{2^m-1}$ are pairwise independent random variables.

Example 3.18. (Generating Pairwise Independent Variables using $\mathbb{F}_2^2$) The non-zero vectors in $\mathbb{F}_2^2$ are $v_1 = (0, 1)$, $v_2 = (1, 0)$, and $v_3 = (1, 1)$. These vectors are pairwise linearly independent.

Let $Y \in \mathbb{F}_2^2$ be chosen uniformly at random. Define $(X_1, X_2, X_3) = (\langle Y, v_1 \rangle, \langle Y, v_2 \rangle, \langle Y, v_3 \rangle)$. By **Remark 3.17**, X_1, X_2, X_3 are pairwise independent random variables. Indeed, if we compute the value of X_1, X_2, X_3 , we get

$Y \in \mathbb{F}_2^2$	X_1	X_2	X_3
(0, 0)	0	0	0
(0, 1)	1	0	1
(1, 0)	0	1	1
(1, 1)	1	1	0

By inspection, each two variables X_i, X_j with $i \neq j$ are independent. Every column has $\frac{1}{2}$ in probability of having 0's or 1's. Every two columns has $\frac{1}{4}$ in probability of having (0, 0)/(0, 1)/(1, 0)/(1, 1).

Example 3.19. (Generating Pairwise Independent Variables using $\mathbb{F}_2^3$) The non-zero vectors in $\mathbb{F}_2^3$ are $v_1 = (0, 0, 1), v_2 = (0, 1, 0), v_3 = (0, 1, 1), v_4 = (1, 0, 0), v_5 = (1, 0, 1), v_6 = (1, 1, 0)$ and $v_7 = (1, 1, 1)$. These vectors are pairwise linearly independent.

Let $Y \in \mathbb{F}_2^3$ be chosen uniformly at random. Define $(X_1, \dots, X_7) = (\langle Y, v_1 \rangle, \dots, \langle Y, v_7 \rangle)$. By **Remark 3.17**, $X_1, \dots, X_7$ are pairwise independent random variables. Indeed, if we compute the value of $X_1, \dots, X_7$, we get

$Y \in \mathbb{F}_2^3$	X_1	X_2	X_3	X_4	X_5	X_6	X_7
(0, 0, 0)	0	0	0	0	0	0	0
(0, 0, 1)	1	0	1	0	1	0	1
(0, 1, 0)	0	1	1	0	0	1	1
(0, 1, 1)	1	1	0	0	1	1	0
(1, 0, 0)	0	0	0	1	1	1	1
(1, 0, 1)	1	0	1	1	0	1	0
(1, 1, 0)	0	1	1	1	1	0	0
(1, 1, 1)	1	1	0	1	0	0	1

By inspection, each two variables X_i, X_j with $i \neq j$ are independent. Every column has $\frac{1}{2}$ in probability of having 0's or 1's. Every two columns has $\frac{1}{4}$ in probability of having (0, 0)/(0, 1)/(1, 0)/(1, 1).

Remark 3.20. We can see in **Example 3.19**, while there are 128 elements in the space $\mathbb{F}_2^7$ there are only 8 are in the support.

Even so, pairwise independence is maintained. This shows random variables a weaker independence can be generated by a random variable with much less “randomness”. For a fully independent distribution we would need all 128 elements.

Meanwhile, the largest collection of 3-wise linearly independent vectors that can generated by $Y \in \mathbb{F}_p^m$ is an area for further exploration.

Corollary 3.21. *To generate n pairwise independent random bits, only $\lceil \log_2(n+1) \rceil$ random bits (for Y) are needed.*

3.2. Chernoff-Bernstein-Hoeffding Bound.

Theorem 3.22. (Chernoff-Bernstein-Hoeffding) *Suppose $X_1, \dots, X_n$ be independent $0, 1$ valued where*

$$\begin{cases} \Pr[X_i = 1] = p \\ \Pr[X_i = 0] = 1 - p \end{cases}$$

Let $X = \sum_{i=1}^n X_i$, we have for arbitrary $t \geq 0$

$$\Pr[|X - np| > t] \leq e^{-\frac{t^2}{3n}}$$

If we choose $\varepsilon = \frac{t}{n}$, we have

$$\Pr[|X - np| > \varepsilon n] \leq e^{-\frac{\varepsilon^2 n}{3}}$$

Proof. (Sketch) Consider for each $i = 1, \dots, n$, define random variables $Z_i = X_i - p$, define Z as

$$Z = \sum_{i=1}^n Z_i = X - np$$

Note that

$$e^{-\frac{t^2}{3n}} \geq \Pr[|X - np| > t] = \Pr[Z > t \vee Z < -t] = \Pr[Z > t] + \Pr[Z < -t]$$

Let arbitrary $c > 0$, we get

$$\begin{aligned} \Pr[Z > t] &= \Pr[cZ > ct] \\ &= \Pr[e^{cZ} > e^{ct}] \\ &= \Pr\left[\prod_{i=1}^n e^{cZ_i} > e^{ct}\right] \\ &\leq \frac{E\left[\prod_{i=1}^n e^{cZ_i}\right]}{e^{ct}} \quad [\text{By Markov}] \\ &= \prod_{i=1}^n \frac{E[e^{cZ_i}]}{e^{ct}} \quad [\text{By full independence}] \\ &= \frac{(p \cdot e^{c(1-p)} + (1-p) \cdot e^{-cp})^n}{e^{ct}} \end{aligned}$$

We first consider the case where $p = \frac{1}{2}$. For the numerator, we get a function with respect to c

$$\begin{aligned} p \cdot e^{c(1-p)} + (1-p) \cdot e^{-cp} &= \frac{e^{\frac{c}{2}} + e^{-\frac{c}{2}}}{2} \quad [\text{This is the cosh function}] \\ &= \frac{\sum_{i=0}^{\infty} \frac{1}{i!} \left(\frac{c}{2}\right)^i + \sum_{i=0}^{\infty} \frac{1}{i!} \left(-\frac{c}{2}\right)^i}{2} \quad [\text{Taylor Expansion}] \\ &= \sum_{i=0}^{\infty} \frac{1}{(2i)!} \left(\frac{c}{2}\right)^{2i} \\ &\leq \sum_{i=0}^{\infty} \frac{1}{i!} \left(\frac{c^2}{2}\right)^i \\ &= e^{\frac{c^2}{2}} \end{aligned}$$

Therefore, we get

$$\Pr[Z > t] \leq \frac{\left(e^{\frac{c^2}{2}}\right)^n}{e^{ct}} = e^{\frac{c^2 n}{2} - ct}$$

We can minimise $e^{\frac{c^2 n}{2} - ct}$ to reach a tighter bound. This value is minimised when $\frac{c^2 n}{2} - ct$ is minimised. We get the value is minimised when $c = \frac{t}{n}$. Since $\frac{t}{n}$ is positive, we get

$$\Pr[Z > t] \leq e^{\frac{t^2}{2n} - \frac{t^2}{n}} = e^{-\frac{t^2}{2n}} \leq e^{-\frac{t^2}{3n}}$$

Now, we consider $p \leq \frac{1}{2}$. We have two cases.

Case 1: Suppose $t \leq 2pn$, we have

$$\begin{aligned} \Pr[Z > t] &\leq \frac{(p \cdot e^{c \cdot (1-p)} + (1-p) \cdot e^{-cp})^n}{e^{ct}} \\ &= \frac{(e^{-cp}(1 + p(e^c - 1)))^n}{e^{ct}} \\ &\leq \frac{(e^{-cp}(1 + p(c + cp)))^n}{e^{ct}} \quad [e^c \leq 1 + c + c^2 \text{ suppose we restrict } c \in [0, 1]] \\ &= \frac{(e^{-cp}(1 + cp + c^2 p))^n}{e^{ct}} \\ &\leq \frac{(e^{-cp} \cdot e^{cp + c^2 p})^n}{e^{ct}} \quad [\forall x. 1 + x \leq e^x] \\ &= e^{c^2 pn - ct} \end{aligned}$$

Choose c to minimise $c^2 pn - ct$ we get $c = \frac{t}{2pn} \implies c \in [0, 1]$ (therefore, we can use the fact that $e^c \leq 1 + c + c^2$). This therefore gives us

$$\Pr[Z > t] = e^{-\frac{t^2}{4pn}} \leq e^{-\frac{t^2}{2n}}$$

similar as the case for $p = \frac{1}{2}$.

Case 2: Suppose $t > 2pn$, we have

$$\Pr[Z > t] \leq \Pr[Z > 2pn] \leq e^{-\frac{4p^2 n^2}{4pn}} \leq e^{-pn}$$

For the case where $Z < -t$, the proof is symmetrical proof. Similarly, for $p \geq \frac{1}{2}$ as well. $\square$

Corollary 3.23. For some function $f \in \omega(\sqrt{n})$, $g \in \Omega(n)$, we have

$$\Pr[|X - np| > f(n)] \leq o(1)$$

$$\Pr[|X - np| > g(n)] \leq e^{-\Omega(n)}$$

3.3. Erdős Rényi Random Graph.

We will now consider a classic example in the probabilistic method. The problem considers a graph of n vertices, where each possible edge exists independently in the graph with some fixed probability p . We want to see how many set of three vertices such that there are edges between each two of them (that is, how many triangles are there in the graph).

Definition 3.24. Given some constants $n \in \mathbb{N}$, $p \in [0, 1]$, a graph $G(n, p) = (V, E)$ contains vertices $V = [n]$, and for each $\{i, j\} \in \binom{[n]}{2}$, $(i, j) \in E$ with a probability p .

Note: p can be constants depended on other variables, like n .

Question 3.25. What is the distribution of the number of triangles in $G(n, p)$

Example 3.26. (Erdős Rényi Random Graph) For each $\{i, j\} \in \binom{[n]}{2}$, define the random variable $Z_{i,j}$ as

$$Z_{ij} = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

For each $\{i, j, k\} \in \binom{[n]}{3}$, define the random variable $X_{i,j,k}$ as

$$X_{i,j,k} = Z_{i,j}Z_{i,k}Z_{j,k} = \begin{cases} 1 & \text{if } (i, j), (j, k), (k, i) \in E \\ 0 & \text{otherwise} \end{cases}$$

We also define the random variable X as the sum of the possible $X_{i,j,k}$'s.

$$X = \sum_{\{i,j,k\} \in \binom{[n]}{3}} X_{i,j,k}$$

We can find the expectation of X

$$\begin{aligned} E[X] &= E \left[\sum_{\{i,j,k\} \in \binom{[n]}{3}} X_{i,j,k} \right] \\ &= \sum_{\{i,j,k\} \in \binom{[n]}{3}} E[X_{i,j,k}] \\ &= \sum_{\{i,j,k\} \in \binom{[n]}{3}} E[Z_{i,j}Z_{i,k}Z_{j,k}] \\ &= \sum_{\{i,j,k\} \in \binom{[n]}{3}} E[Z_{i,j}]E[Z_{i,k}]E[Z_{j,k}] \quad [\text{By independence}] \\ &= p^3 \binom{n}{3} \end{aligned}$$

We can also find the variance of X .

$$\begin{aligned} \text{Var}(X) &= E \left[\left(X - p^3 \binom{n}{3} \right)^2 \right] \\ &= E \left[\left(\sum_{\{i,j,k\} \in \binom{[n]}{3}} (X_{i,j,k} - p^3) \right)^2 \right] \\ &= E \left[\sum_{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3}} (X_{i,j,k} - p^3)(X_{i',j',k'} - p^3) \right] \\ &= \sum_{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \\ &= \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| \leq 1}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \\ &\quad + \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| = 2}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \\ &\quad + \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| = 3}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \end{aligned}$$

We will deal with these sums separately. For the first sum, since $|\{i, j, k\} \cap \{i', j', k'\}| \leq 1$, having at most one vertex being shared means that all 6 edges are generated independently. Hence, $X_{i,j,k}, X_{i',j',k'}$ which are generated by independent set of random variables are independent. And we get

$$\begin{aligned}
& \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| \leq 1}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \\
&= \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| \leq 1}} E[(X_{i,j,k} - p^3)] E[X_{i',j',k'} - p^3] \\
&= \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| \leq 1}} (p^3 - p^3)(p^3 - p^3) \\
&= 0
\end{aligned}$$

Now for the second sum, since $|\{i,j,k\} \cap \{i',j',k'\}| = 2$, having two vertices being shared means that the two triangles is made of only five distinct edges

$$\begin{aligned}
& \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| = 2}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \\
&= \sum_{\{a,b,c,d\} \in \binom{[n]}{4}} E[(X_{a,b,c} - p^3)(X_{a,b,d} - p^3)] \\
&= \sum_{\{a,b,c,d\} \in \binom{[n]}{4}} (E[X_{a,b,c}X_{a,b,d}] - E[p^3X_{a,b,c}] - E[p^3X_{a,b,d}] + E[p^6]) \\
&= \sum_{\{a,b,c,d\} \in \binom{[n]}{4}} \left(\sum_{c \in \{0,1\}} \Pr[X_{a,b,c}X_{a,b,d} = c]c - p^3E[X_{a,b,c}] - p^3E[X_{a,b,d}] + E[p^6] \right) \\
&= \sum_{\{a,b,c,d\} \in \binom{[n]}{4}} (p^5 - p^6) \quad [\Pr[X_{a,b,c}X_{a,b,d} = 1] = p^5 \text{ as the two triangles has only five edges}] \\
&= \Theta(n^4)(p^5 - p^6)
\end{aligned}$$

Finally, for the third sum, since $|\{i,j,k\} \cap \{i',j',k'\}| = 3$, having three vertices being shared means that two triangles are the same, and is made of only three distinct edges

$$\begin{aligned}
& \sum_{\substack{\{i,j,k\}, \{i',j',k'\} \in \binom{[n]}{3} \\ |\{i,j,k\} \cap \{i',j',k'\}| = 3}} E[(X_{i,j,k} - p^3)(X_{i',j',k'} - p^3)] \\
&= \sum_{\{i,j,k\} \in \binom{[n]}{3}} E[X_{i,j,k}^2] - E[p^3X_{i,j,k}] - E[X_{i,j,k}p^3] + E[p^6] \\
&= \sum_{\{i,j,k\} \in \binom{[n]}{3}} \left(\sum_{c \in \{0,1\}} \Pr[X_{i,j,k}^2 = c]c - p^3E[X_{i,j,k}] - p^3E[X_{i,j,k}] + E[p^6] \right) \\
&= \sum_{\{i,j,k\} \in \binom{[n]}{3}} (p^3 - p^6) \\
&= \Theta(n^3)(p^3 - p^6)
\end{aligned}$$

Add the above results up, we get

$$\text{Var}[X] = \Theta(n^4)(p^5 - p^6) + \Theta(n^3)(p^3 - p^6)$$

Remark 3.27. (Erdős Rényi Random Graph with fixed probability) Suppose $p = \frac{1}{2}$, where the graph $G(n, p)$ generates edges uniformly random, we get

$$\Pr \left[\left| X - \frac{1}{8} \binom{n}{3} \right| > t \right] \leq \mathcal{O} \left(\frac{n^4}{t^2} \right)$$

So if $t = \omega(n^2)$ as $\mathcal{O} \left(\frac{n^4}{t^2} \right) = o(1)$, we have

$$\Pr \left[\left| X - \frac{1}{8} \binom{n}{3} \right| > t \right] \leq o(1) \stackrel{\text{Equivalently}}{\iff} \Pr \left[X \in \left[\frac{1}{8} \binom{n}{3} - \omega(n^2), \frac{1}{8} \binom{n}{3} + \omega(n^2) \right] \right] = 1 - o(1)$$

Note: Similar results works for arbitrary p , we are just setting $p = \frac{1}{2}$ as an example.

Remark 3.28. (Erdős Rényi Random Graph with variable probability)

1. To answer which $p = p(n)$ can we have that

$$\Pr[X > 0] = 1 - o(1)$$

We have $X = 0$ means $|X - \binom{n}{3}p^3| \geq \binom{n}{3}p^3$. So we pick $t = \binom{n}{3}p^3$. We have

$$\begin{aligned} \Pr[X = 0] &= \Pr \left[\left| X - \binom{n}{3}p^3 \right| \geq \binom{n}{3}p^3 \right] \\ &\leq \mathcal{O} \left(\frac{n^4 p^5 + n^3 p^3}{n^6 p^6} \right) \quad [\text{By Chebyshev}] \\ &= \mathcal{O} \left(\frac{1}{n^2 p} + \frac{1}{n^3 p^3} \right) \end{aligned}$$

Therefore, if we set $p = \omega(\frac{1}{n})$ then $\frac{1}{p} = o(n^2)$ and $p = o(n)$. Therefore,

$$\mathcal{O} \left(\frac{1}{n^2 p} + \frac{1}{n^3 p^3} \right) = o(1) \implies \Pr[X = 0] = o(1)$$

2. To answer which $p = p(n)$ can we have that

$$\Pr[X = 0] = 1 - o(1)$$

If we set $p = o(\frac{1}{n})$ then $\frac{1}{p} = \omega(n)$ and therefore,

$$\Pr[X = 0] \leq \mathcal{O} \left(\frac{1}{n^2 p} + \frac{1}{n^3 p^3} \right) = \omega(1) \implies \Pr[X = 0] = 1 - o(1)$$

Another way of seeing this is by Markov's Inequality

$$\Pr[X = 0] = 1 - \Pr[X > 0.5] \leq \frac{E[X]}{0.5} = \frac{p^3 \binom{n}{3}}{0.5} = 1 - o(1)$$

3.4. Maximum Cut.

Definition 3.29. (Max Cut) Given an undirected graph $G = (V, E)$. A **cut** is a partition of $V = S \sqcup S^c$.

We define the **max cut** of G , denoted by $\text{MAXCUT}(G)$ is

$$\text{MAXCUT}(G) = \max_{S \subseteq V} \{|e(S, S^c)|\} = \max_{S \subseteq V} \{|\{(u, v) \in E : u \in S, v \in S^c\}|\}$$

is maximised.

From now on, we will focus on the following question

Question 3.30. Given a graph G , is there an algorithm that finds the max cut of G .

Example 3.31. In a bipartite graph $G = (L, R, E)$, the cut $V = L \sqcup R$ is the max cut. In this case

$$\text{MAXCUT}(G) = |E|$$

Hence, to find the max cut of certain graphs, we can check a graph is bipartite or not. If it is, we can immediately return $|E|$ as the max cut. The following algorithm checks whether a given graph G is bipartite or not,

- Take arbitrary $v \in V$ and put it in S
- Put all of the neighbours of v in S^c
- For each neighbour, put its neighbours into S
- Repeat until finding a vertex that is placed in both S and S^c . In this case, return the graph is not bipartite. Otherwise, return the max cut S, S^c .

The algorithm runs in $\text{poly}(|V|)$ time.

Remark 3.32. The problem of finding $\text{MAXCUT}(G)$ is NP-hard. Although finding the exact value might be really hard, we can try to find approximation algorithm for it

Definition 3.33. An algorithm A is an α -approximation algorithm for MAXCUT if for all graphs G

$$\alpha \cdot \text{MAXCUT} \leq A(G) \leq \text{MAXCUT}$$

We will find a $\frac{1}{2}$ -approximation algorithm for MAXCUT .

Algorithm 3.34. (First attempt: $\frac{1}{2}$ -approximation for MAXCUT) Consider the following algorithm, on input $G = (V, E)$

- For each v , put v into S, S^c independently with probability $\frac{1}{2}$
- Output S, S^c

We can show this is, “on expectation”, a $\frac{1}{2}$ -approximation for MAXCUT

Remark 3.35. For the algorithm A above, for each edge $(u, v) \in E$, denote the random variable $X_{u,v}$ to be

$$X_{u,v} = \begin{cases} 1 & u, v \text{ are in different parts} \\ 0 & \text{otherwise} \end{cases}$$

Note that the $X_{u,v}$'s are not independent, as it is impossible for all of the $X_{u,v}, X_{v,w}, X_{u,w}$ to be 1.

We have

$$\begin{aligned} E_{S, S^c \leftarrow A}[|e(S, S^c)|] &= E_{S, S^c \leftarrow A} \left[\sum_{(u,v) \in E} X_{u,v} \right] \\ &= \sum_{(u,v) \in E} E_{S, S^c \leftarrow A}[X_{u,v}] \\ &= \frac{1}{2} \cdot |E| \\ &\geq \frac{1}{2} \text{MAXCUT}(G) \end{aligned}$$

Therefore, this algorithm is a $\frac{1}{2}$ -approximation “on expectation”, that is

$$\frac{1}{2} \text{MAXCUT}(G) \leq E_{S, S^c \leftarrow A}[|e(S, S^c)|] \leq \text{MAXCUT}(G)$$

Note that **Algorithm 3.34** only leveraged pairwise independence of assigning vertices into one of the two sets.

Indeed, one can easily check that, assuming only pairwise independence, the probability of u, v are separated or at the same set are exactly $\frac{1}{2}$ each. That is, each random variable have exactly probability of $\frac{1}{2}$ to be evaluated to 0 or 1.

Thus, full independence is not necessary. We will now try to use this fact to construct a working algorithm.

Algorithm 3.36. (Second Attempt: $\frac{1}{2}$ -approximation for MAXCUT) Consider the following algorithm, on input $G = (V, E)$

- Denote $|V| = n$, take $t = \lceil \log_2(n+1) \rceil$.
- Index each vertices v by distinct elements of $\mathbb{F}_2^t$.
- For each $a \in \mathbb{F}_2^t$
 - For each $v \in \mathbb{F}_2^t$, assign the vertex v into

$$\begin{cases} S & \text{if } \langle a, v \rangle = 0 \\ S^c & \text{if } \langle a, v \rangle = 1 \end{cases}$$

- Keep the S, S^c with the largest $|e(S, S^c)|$
- Return S, S^c .

Remark 3.37. For the algorithm A above, consider $a \in \mathbb{F}_2^t$ chosen uniformly at random, then we have assigning each vertex into one of the S, S^c is pairwise independent by [Corollary 3.21](#). Hence, we get for all $(u, v) \in E$

$$\Pr[X_{u,v} = 1] = \Pr[\langle a, u \rangle \neq \langle a, v \rangle] = \frac{1}{2} \implies E_{\alpha \leftarrow \mathbb{F}_2^t}[X_{u,v}] = \frac{1}{2}$$

Thus, we have

$$\begin{aligned} E_{\alpha \leftarrow \mathbb{F}_2^t}[|e(S, S^c)|] &= E_{\alpha \leftarrow \mathbb{F}_2^t} \left[\sum_{(u,v) \in E} X_{u,v} \right] \\ &= \sum_{(u,v) \in E} E_{\alpha \leftarrow \mathbb{F}_2^t}[X_{u,v}] \\ &= \frac{1}{2} \cdot |E| \\ &\geq \frac{1}{2} \text{MAXCUT}(G) \end{aligned}$$

Since the algorithm outputs S, S^c with $|e(S, S^c)|$ maximised out of all possible a , we must have the S, S^c output by A satisfies

$$\frac{1}{2} \text{MAXCUT}(G) \leq |e(S, S^c)| \leq \text{MAXCUT}(G)$$

which is indeed a $\frac{1}{2}$ -approximation algorithm.

3.5. Max-3-SAT.

Definition 3.38. (Max-SAT problem) Let $X_1, \dots, X_n$ be variables that can be evaluated as T/F .

Consider m clauses $C_1, \dots, C_m$, where each clause composes with distinct variables, possibly negated. Here is some examples of a clause could be.

- $X_3 \vee X_5 \vee (\neg X_{10})$
- $X_3 \vee X_4 \vee (\neg X_{10}) \vee X_{19}$

Then, we define Max-SAT value is

$$\max_{\text{assignments of } X_1, \dots, X_n} \{ \# \text{ satisfied clauses} \}$$

Definition 3.39. (Max-E3-SAT problem) Let $X_1, \dots, X_n$ be variables that can be evaluated as T/F .

Consider m clauses $C_1, \dots, C_m$, where each clause composes with 3 distinct variables, possibly negated.

Then, we define Max-E3-SAT value is

$$\max_{\text{assignments of } X_1, \dots, X_n} \{ \# \text{ satisfied clauses} \}$$

Algorithm 3.40. ($\frac{7}{8}$ -approximation for Max-E3-SAT) Consider the following algorithm

- For each X_i , randomly assign X_i with a truth value T/F independently with probability $\frac{1}{2}$

We can show this is a $\frac{7}{8}$ -approximation for Max-E3-SAT

Theorem 3.41. *Consider the algorithm A in Algorithm 3.40*

$$\frac{7}{8} \text{Max-E3-SAT}(G) \leq E_{X_1, \dots, X_n \leftarrow A}[\# \text{ satisfied clauses}]$$

Proof. For each clause C_i , denote the random variable Z_i to be

$$Z_i = \begin{cases} 1 & C_i \text{ is satisfied} \\ 0 & \text{otherwise} \end{cases}$$

We have

$$\begin{aligned} E_{X_1, \dots, X_n \leftarrow A}[\# \text{ satisfied clauses}] &= E_{X_1, \dots, X_n \leftarrow A} \left[\sum_{i=1}^m Z_i \right] \\ &= \sum_{i=1}^m E_{X_1, \dots, X_n \leftarrow A}[Z_i] \\ &= \frac{7}{8} \cdot m \\ &\geq \frac{7}{8} \text{Max-E3-SAT}(G) \end{aligned}$$

□

3.6. Ramsay Theory.

The main statement of Ramsay Theory is in the form: in any large enough object, there is some structure somewhere.

For example, the Law of large number, states some results about what happens when the trial number is large enough.

The best starting point for now is the Pigeonhole Principle

Theorem 3.42. (Pigeonhole Principle/PHP)

- (Basic) If we c -colour $c + 1$ items, some 2 have the same colour.
- (Generalised) If we c -colour more than $n(k - 1) + 1$ items, then some k have the same colour.

3.6.1. Ramsay Theorem for Regular Graphs.

Now we proceed with Ramsay Theory, Ramsay Theory is about answering the following

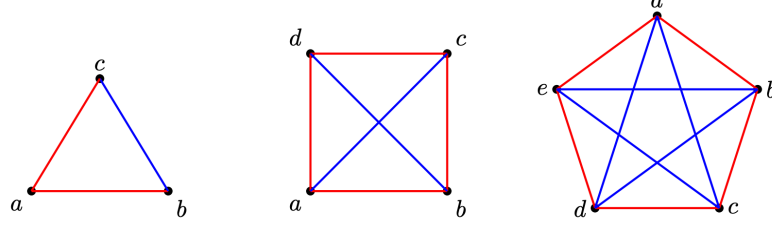
Theorem 3.43. (Ramsay Theorem) *Given some c, k , we are concerning with c -colouring graphs.*

What does $R_c(k)$ need to be such that for all $n \geq R_c(k)$, for all ways of c -colouring all edges of K_n , there is a monochromatic (meaning all the edges have the same colour in the object) K_k .

We will focus on studying the properties of $R_c(k)$. From now on, we will focus on the case where $c = 2$.

Example 3.44. We will show how to find $R_c(k)$ for $k = 2, 3$

- For $k = 2$, K_2 is just an edge. Hence, we need $n \geq 2$ to make sure that we have a monochromatic K_2 . Therefore $R_2(2) = 2$.
- For $k = 3$, we can produce a counter example where there is a colouring with no K_k for each of the $n = 3, 4, 5$ case as follows



However, $n = 6$ works, take any vertex $v \in K_6$, it has 5 edges. By PHP, at least 3 have the same colour. WLOG, assume the colour is red, and call the 3 vertices v_1, v_2, v_3 . Then the edges $(v, v_1), (v, v_2), (v, v_3)$ are all red.

Now, if there is some $i, j \in [3], i \neq j$ where (v_i, v_j) is red, then v, v_i, v_j is a monochromatic K_3 graph that is coloured in red.

Otherwise, all (v_i, v_j) where $i, j \in [3], i \neq j$ is coloured in blue. And v_1, v_2, v_3 is a monochromatic K_3 graph that is coloured in blue.

Therefore $R_2(3) = 6$.

Now, one might try to generalise a lowerbound for $R_2(k)$.

Our **first attempt** would be using induction.

Base case: We have covered the case for $k = 2$ and $k = 3$.

Inductive step: Suppose we have done for $R_2(2), \dots, R_2(k)$ and want to find $R_2(k+1)$. One might attempt to do the following:

Take any $n \geq R_2(k+1)$, for any arbitrary vertex v , by PHP, there are at least $\lceil \frac{n-1}{2} \rceil$ vertices such that the edge from one of those vertex and v have the same colour. If we take n to satisfy $\lceil \frac{n-1}{2} \rceil \geq R(k)$ then we can use the Inductive Hypothesis on those $\lceil \frac{n-1}{2} \rceil$ vertices. And we can conclude that there is a monochromatic K_k .

However, we never know if the color for the monochromatic K_k aligns with the color of edges from the vertex v .

Therefore, we now introduces a new definition that will solve the problem

Definition 3.45. For any k, l , denote $R(k, l)$ to be the number that for all $n \geq R(k, l)$, for all ways of 2-colouring all edges of K_n , there is always a red K_k or a blue K_l .

Remark 3.46. Note that we have the following properties

- $\forall k, R(k, k) = R_2(k)$
- $\forall k, l, R(k, l) = R(l, k)$

We will now prove a bound of $R(k, l)$

Theorem 3.47. For all $k, l, R(k, l) \geq 2 \max\{R(k-1, l), R(k, l-1)\}$ is enough to satisfy the properties of R .

Proof. **Base case:** $\forall k, l, R(k, 2) = k, R(2, l) = l$ is vacuously true. We can easily find those values as we either have entire K_k is coloured by one colour, which case we have a monochromatic K_k in this colour. Or if there is the other colour, then we got a monochromatic K_2 in the other colour.)

Inductive step: We will show $R(k, l) = 2 \max\{R(k-1, l), R(k, l-1)\}$ is enough. Consider arbitrary vertex v , by PHP, there are

$$\max\{R(k-1, l), R(k, l-1)\}$$

edges that connects to v that is in the same colour. Consider the subgraph induced by those $\max\{R(k-1, l), R(k, l-1)\}$ vertices that connects v with those $\max\{R(k-1, l), R(k, l-1)\}$ edges.

If $R(k-1, l) \geq R(k, l-1)$, by IH, this subgraph has either a monochromatic K_l that is in blue, or it has a monochromatic K_{k-1} that is in red. We will argue those in cases

- If the subgraph has a monochromatic K_l in blue, we are done

- If the subgraph has a monochromatic K_{k-1} in red.
 - If all $\max\{R(k-1, l), R(k, l-1)\}$ edges are coloured in red. Then, together with v we get a monochromatic K_k that is in red.
 - If all $\max\{R(k-1, l), R(k, l-1)\}$ edges are coloured in blue. Then since $\max\{R(k-1, l), R(k, l-1)\} > R(k, l-1)$. The subgraph either have a monochromatic K_k in red or monochromatic K_{l-1} in blue
 - If we have a monochromatic K_k in red, we are done.
 - If we have a monochromatic K_{l-1} in blue, then together with v we get a monochromatic K_l in blue.

The case for $R(k-1, l) < R(k, l-1)$ is similar. If $R(k-1, l) < R(k, l-1)$, by IH, this subgraph has either a monochromatic K_k that is in red, or it has a monochromatic K_{l-1} that is in blue. We will argue those in cases

- If the subgraph has a monochromatic K_k in red, we are done
- If the subgraph has a monochromatic K_{l-1} in blue.
 - If all $\max\{R(k-1, l), R(k, l-1)\}$ edges are coloured in blue. Then, together with v we get a monochromatic K_l that is in blue.
 - If all $\max\{R(k-1, l), R(k, l-1)\}$ edges are coloured in red. Then since $\max\{R(k-1, l), R(k, l-1)\} > R(k-1, l)$. The subgraph either have a monochromatic K_{k-1} in red or monochromatic K_l in blue
 - If we have a monochromatic K_l in blue, we are done.
 - If we have a monochromatic K_{k-1} in red, then together with v we get a monochromatic K_k in red.

□

Remark 3.48. The proof of the Ramsey theory for the general c -colouring is similar to above.

Theorem 3.49. For all k, l

$$R(k, l) \leq R(k-1, l) + R(k, l-1)$$

Proof. Base case: $\forall k, l, R(k, 2) = k, R(2, l) = l$ is vacuously true. We can easily find those values as we either have entire K_k is coloured by one colour, which case we have a monochromatic K_k in this colour. Or if there is the other colour, then we got a monochromatic K_2 in the other colour.)

Inductive step: We will show $R(k, l) = R(k-1, l) + R(k, l-1)$ is enough. Consider arbitrary vertex v , take $R = \{v' \in V : (v, v') \text{ has colour red}\}, B = \{v' \in V : (v, v') \text{ has colour blue}\}$.

By PHP, either $|R| \geq R(k-1, l)$ or $|B| \geq R(k, l-1)$.

- If $|R| \geq R(k-1, l)$. If the subgraph induced by $R \cup \{v\}$ has a blue K_l , we are done. Otherwise, the subgraph induced by R has a red K_{k-1} and $R \cup \{v\}$ has a red K_k .
- If $|B| \geq R(k, l-1)$. If the subgraph induced by $B \cup \{v\}$ has a red K_k , we are done. Otherwise, the subgraph induced by B has a blue K_{l-1} and $B \cup \{v\}$ has a blue K_l .

□

Theorem 3.50. (Ramsay Bound for Ramsay Theorem) $R(k, k) \leq 2^{2k} = 4^k \in \mathcal{O}(4^k)$

Proof. We will show in general that $R(k, l) \leq 2^{k+l-3}$

Base case: For any $k \geq 2$, we have $R(k, 2) = k = R(2, k) < 2^{k+2-3} = 2^{k-1}$

Inductive step: We have

$$\begin{aligned}
R(k+1, l) &= 2 \max\{R(k, l), R(k+1, l-1)\} \\
&= 2 \cdot 2^{k+l-3} \quad [\text{By IH}] \\
&= 2^{(k+1)+l-3}
\end{aligned}$$

We also have

$$\begin{aligned}
R(k, l+1) &= 2 \max\{R(k-1, l+1), R(k, l)\} \\
&= 2 \cdot 2^{k+l-3} \quad [\text{By IH}] \\
&= 2^{(k+1)+l-3}
\end{aligned}$$

Hence, this gives

$$R(k, k) \leq 2^{2k-3} \in \mathcal{O}(4^k)$$

□

Theorem 3.51. (Erdős) *We cannot take $R(k, k) \leq o(k \cdot 2^{\frac{k}{2}})$*

Proof. (By Probabilistic method) Given k and the corresponding $n = R(k, k) \leq o(k \cdot 2^{\frac{k}{2}})$. We show there is a 2-coloring of K_n with no monochromatic K_k .

Suppose we colour each edge with red/blue with $\frac{1}{2}$ probability independently. We want to show

$$\Pr[\exists \text{ a monochromatic } K_k] < 1$$

Fix any $S \subseteq [n]$ where $|S| = k$. Let E_S be the event that K_S is monochromatic. Then

$$\Pr[E_S] = 2 \cdot \frac{1}{2^{\binom{k}{2}}}$$

Thus

$$\Pr[\exists \text{ a monochromatic } K_k] = \Pr\left[\bigcup_{S \in \binom{[n]}{k}} E_S\right] \leq \sum_{S \in \binom{[n]}{k}} \Pr[E_S] \leq \binom{n}{k} \frac{2}{2^{\binom{k}{2}}}$$

The next step is to use the following fact

Fact:

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k$$

Therefore,

$$\Pr[\exists \text{ a monochromatic } K_k] \leq \frac{2^{\binom{n}{k}}}{2^{\binom{k}{2}}} \leq \frac{2^{\left(\frac{en}{k}\right)^k}}{2^{\binom{k}{2}}} = 2^{k \log\left(\frac{en}{k}\right) - \binom{k}{2} + 1}$$

We want the probability to be less than 1 so $k \log\left(\frac{en}{k}\right) - \binom{k}{2} + 1 < 0$. This gives

$$\binom{k}{2} > k \log\left(\frac{en}{k}\right) + 1 \implies \frac{k-1}{2} + \log(k) - \frac{1}{k} > \log(n) + \log(e)$$

And we can easy to see that when $n \in o(k \cdot 2^{\frac{k}{2}})$, the above inequality holds. □

Remark 3.52. In fact, we can show that $k \log\left(\frac{en}{k}\right) - \binom{k}{2} + 1 < -99$, we get

$$\binom{k}{2} > k \log\left(\frac{en}{k}\right) + 100 \implies \frac{k-1}{2} + \log(k) - \frac{100}{k} > \log(n) + \log(e)$$

This shows that $\Pr[\exists \text{ a monochromatic } K_k] < 2^{-99}$

Theorem 3.53. (Infinite Ramsey Theorem) *For any 2-colouring of K_N there exists an infinite $S \subseteq \mathbb{N}$ such that K_S is monochromatic.*

Proof. Take any vertex $v_0 \in \mathbb{N}$, there are infinitely many edges connected to v_0 . By PHP, there is an infinite set $S_0 \subseteq \mathbb{N}$ such that all (v_0, v) where $v \in S_0$ are coloured by the same colour $c_0 \in \{R, B\}$. Pick such S_0 .

Now, continue to do the following. For all $i \geq 0$, pick $v_{i+1} \in S_i$, by PHP, there is an infinite set $S_{i+1} \subseteq S_i$ such that all (v_{i+1}, v) where $v \in S_{i+1}$ are coloured by the same colour $c_{i+1} \in \{R, B\}$. Pick such S_{i+1} .

This gives us a sequence of $v_0, v_1, \dots$, a sequence of $S_0 \supseteq S_1 \supseteq \dots$, and a sequence of $c_0, c_1, \dots$ that satisfies the following

- $\forall i \geq 0, v_{i+1} \in S_i$
- For all edges (v_i, v) where $v \in S_i$ has colour c_i .
- By the previous claim, we have for all $i < j$, (v_i, v_j) has colour c_i (since $v_j \in S_{j-1} \subseteq S_i$).

By PHP, there exists a colour $c^* \in \{R, B\}$ occurred infinitely many times in $c_0, c_1, \dots$. That is, there is an infinite set $J \subseteq \mathbb{N}$ such that $c_j = c^*$ for all $j \in J$.

Consider $S = \{v_j : j \in J\}$, for all $j_1, j_2 \in J$, the colour of (v_{j_1}, v_{j_2}) is $c_{\min(j_1, j_2)} = c^*$. $\square$

3.6.2. Ramsey Theorem for Hypergraphs.

Definition 3.54. Consider V as a finite vertex set

- A 3-uniform hypergraph is a graph (V, E) where $E \subseteq \binom{V}{3}$ (each edge connects 3 vertices)
- The complete 3-uniform hypergraph on a set V is the hypergraph $(V, \binom{V}{3})$, denoted as $K_V^{(3)}$.

Theorem 3.55. (3-uniform Hypergraph Ramsey Theorem) *For all k , there exists a number $n_0(k)$, such that for all $n \geq n_0(k)$, for any 2-colouring of $K_{[n]}^{(3)}$, there exists a set $S \subseteq [n], |S| = k$ such that $K_S^{(3)}$ is monochromatic.*

There are two ways to prove this, we will present both of them

Proof. (By applying Ramsey Theorem on Regular Graphs many times, then PHP)

We will fix n from now, take any vertex v_0 , there are $\binom{n-1}{2}$ hyperedges connected to v_0 .

Consider a regular graph $K_{[n] \setminus \{v_0\}}$, consider the 2-colouring of this graph that for each edge (a, b) , we use the colour of the hyperedge of $\{v_0, a, b\}$ in the hypergraph. By Ramsey Theorem for Regular Graphs, there exists a set $S_0 \subseteq [n] \setminus \{v_0\}$ such that all $a, b \in S_0$, (a, b) have the same colour c_0 .

Now, do the following for $2k - 1$ more times. For all $i \geq 0$, pick $v_{i+1} \in S_i$. Consider a regular graph $K_{S_i \setminus \{v_{i+1}\}}$ that for each edge (a, b) , we use the colour of the hyperedge of $\{v_{i+1}, a, b\}$ in the hypergraph. By Ramsey Theorem for Regular Graphs, there exists a set $S_{i+1} \subseteq S_i \setminus \{v_{i+1}\}$ such that all $a, b \in S_{i+1}$, (a, b) have the same colour c_{i+1} .

This gives us a sequence of $v_0, v_1, \dots, v_{2k-1}$, a sequence of $S_0 \supseteq S_1 \supseteq \dots \supseteq S_{2k-1}$, and a sequence of $c_0, c_1, \dots, c_{2k-1}$ that satisfies the following

- $\forall i \geq 0, v_{i+1} \in S_i$
- For all $a, b \in S_i$, the hyperedge $\{v_i, a, b\}$ has colour c_i .
- By the previous claim, we have for all $a < b < c$, (v_a, v_b, v_c) has colour c_a (since $v_c \in S_{c-1} \subseteq S_a \supseteq S_{b-1} \ni v_b$).

By PHP, there exists a colour $c^* \in \{R, B\}$ occurred at least k times in $c_0, c_1, \dots, c_{2k-1}$. That is, there is a set $J \subseteq \{0, 1, \dots, 2k-1\}$ such that $c_j = c^*$ for all $j \in J$.

Consider $S = \{v_j : j \in J\}$, for all $j_1, j_2, j_3 \in J$, the colour of $\{v_{j_1}, v_{j_2}, v_{j_3}\}$ is $c_{\min(j_1, j_2, j_3)} = c^*$.

In particular, for each S_i , we apply Ramsey Theorem one time on this set to obtain S_{i+1} . We get $|S_{i+1}| \in \Omega(\log_4(|S_i|))$ (Indeed, for S_{i+1} to be size k , we need S_i to be size $\mathcal{O}(4^k)$). Thus, for simplicity, we will assume $|S_{i+1}| = \log_4(|S_i|)$ as the answers are the same up to some factors). Therefore, in order to make sure S_{2k-1} is not empty so that we can pick some arbitrary v in this set, we require applying $\log_4$ on n for $2k$ times gives us at least 1. Hence we get

$$n \geq 4^{\cdot 2k \text{ times} \cdot 4}$$

Hence, we can set $n_0(k) = 4^{\cdot 2k \text{ times} \cdot 4}$. $\square$

Proof. (By applying PHP many times, then Ramsey Theorem for Regular Graphs)

Let $\chi : \binom{V}{3} \rightarrow \{R, B\}$ be the colouring.

Take any vertices v_0, v_1 there are $n - 2$ hyperedges $\{v_0, v_1, a\}$. By PHP, at least $\frac{n-2}{2}$ edges have the same colour c_{01} . Let $S_1 \subseteq [n] \setminus \{v_0, v_1\}$ be the set such that for all $a \in S_1$, $\chi(\{v_0, v_1, a\}) = c_{01}$.

Now, do the following for $m - 1$ more times. For all $i \geq 0$, pick any $v_{i+1} \in S_i$, for each $a \in S_i \setminus \{v_{i+1}\}$, consider the colour profile $(\chi(\{v_0, v_{i+1}, a\}), \chi(\{v_1, v_{i+1}, a\}), \dots, \chi(\{v_i, v_{i+1}, a\})) \in \{R, B\}^i$. There are 2^{i+1} possible colour profiles, so by PHP, some colour profile has at least $\frac{|S_i|-1}{2^{i+1}}$, we will denote this colour profile as $(c_{0(i+1)}, c_{1(i+1)}, \dots, c_{i(i+1)})$. Define $S_{i+1} \subseteq S_i$

Define

$$S_{i+1} = \left\{ a \in S_i \setminus \{v_{i+1}\} : (\chi(\{v_0, v_{i+1}, a\}), \chi(\{v_1, v_{i+1}, a\}), \dots, \chi(\{v_i, v_{i+1}, a\})) = (c_{0(i+1)}, c_{1(i+1)}, \dots, c_{i(i+1)}) \right\}$$

This gives us a sequence of $v_0, v_1, \dots, v_m$, a sequence of $S_0 \supseteq S_1 \supseteq \dots, S_m$, and $c_{ij}, 0 \leq i < j \leq m$ that satisfies the following

- $\forall i \geq 1, v_{i+1} \in S_i$
- $\forall i \geq 0, |S_{i+1}| \geq \frac{|S_i|-1}{2^{i+1}}$
- For arbitrary v_i , take any $j < k < i$, then $\chi(\{v_j, v_k, v_i\}) = c_{jk}$ (since $v_i \in S_{i-1} \subseteq S_k$)

Now, consider the graph $K_{\{v_1, \dots, v_m\}}$ where each edge (v_i, v_j) (assuming $i < j$) is coloured in c_{ij} . By Ramsey Theorem for Regular Graphs, if $m \geq 4^k$, there exists a monochromatic K_k in $K_{\{v_1, \dots, v_m\}}$ with the colour c^* . That is, there is a set $J \subseteq \{v_1, \dots, v_m\}$ where $|J| = k$ such that for all $v_i, v_j \in J$ (assuming $i < j$)

$$c_{ij} = c^*$$

Then, for all $v_i, v_j, v_k \in J$ (assuming $i < j < k$)

$$\chi(\{v_i, v_j, v_k\}) = c_{ij} = c^*$$

showing that J is the set where $K_J^{(3)}$ in the original hypergraph is monochromatic.

Now, we already have $m \geq 4^k$. Moreover, in order for S_m to be non-empty.

$$n \cdot \prod_{i=1}^m \frac{1}{2^i} \geq 1 \implies \frac{n}{2^{\frac{m(m+1)}{2}}} \geq 1 \implies n \geq 2^{\frac{m(m+1)}{2}} = 2^{\frac{4^{2k}+4^k}{2}}$$

Hence, we can set $n_0(k) = 2^{\frac{4^{2k}+4^k}{2}} \approx 2^{4^{2k}}$. This bound is much better than the previous one. $\square$

3.7. Turan Type Theorems.

This section considers the following problem.

Question 3.56. Given some graph H , what is the value of $\text{ex}(n, H)$ such that for any graph G on n vertices, if G contains $> \text{ex}(n, H)$ edges, then it contains a copy of H as a subgraph.

We will first consider the case where $H = K_3$, which is a triangle. We have the following observations

Example 3.57. We can see that for any graph G on n vertices

- $n - 1$ does not guarantee a triangle, e.g. a tree that spans all the vertices. Hence, $f_H(n) > n - 1$.
- n does not guarantee a triangle, e.g. a cycle that spans all the vertices. Hence, $f_H(n) > n$.
- $4n$ does not guarantee a triangle, e.g. arrange the vertices into an $\sqrt{n} \times \sqrt{n}$ grid, and connect them like a grid, this gives around $4n$ edges. Hence, $f_H(n) > 4n$.
- $\binom{n}{2}$ guarantees a triangle, as it contains all the possible edges. Hence, $f_H(n) \leq \binom{n}{2}$.
- $\frac{n^2}{4}$ does not guarantee a triangle, e.g. the complete $(\frac{n}{2}, \frac{n}{2})$ -bipartite graph. Hence, $f_H(n) > \frac{n^2}{4}$.

Note that in the complete $(\frac{n}{2}, \frac{n}{2})$ -bipartite graph, adding another edge would create a triangle. It turns out that this is exactly the case where the maximum number of edges that a graph G can have without creating a triangle.

3.7.1. Turan's Theorem for Cliques.

Theorem 3.58. (Turan's Theorem for K_3) Consider $H = K_3$, then $\text{ex}(n, H) = \lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} \rceil$.

Proof. The idea is essentially reducing this problem from any graphs with n vertices to bipartite graphs with n vertices. This can be done by showing that for any graphs with n vertices with no triangles, we can construct a bipartite graph with at least the same edges. And bipartite graph certainly does not have any triangles.

Let $G = (V, E)$ be an n vertex graph with no triangles.

Let v be a vertex with the maximal degree in G , say the degree is d . Consider the neighbourhood $N(v)$ of v . Let $l = n - |N(v)|$ and $v_1, \dots, v_l$ be the copies of v . Consider a new graph $G' = (V', E')$ defined as the following,

$$\begin{aligned} V' &= N(v) \cup \{v_1, \dots, v_l\} \\ E' &= \{(u, w) : u, w \in N(v) \wedge (u, w) \in E\} \cup \{(u, v_i) : u \in N(v), i \in [l]\} = \{(u, v_i) : u \in N(v), i \in [l]\} \end{aligned}$$

This gives us the following observation

- Note that $\{(u, w) : u, w \in N(v) \wedge (u, w) \in E\} = \emptyset$ as otherwise there will be a triangle in the original graph G . Since there are no edges that connects some elements in $N(v)$ to some other elements in $N(v)$, any edge in G must connects some element in $V \setminus N(v)$ to some element in V . Therefore, we get

$$\begin{aligned} |E| &= |e(V \setminus N(v), V)| \\ &\leq |V \setminus N(v)|d \\ &= |E'| \end{aligned}$$

Hence, G' has at least the same number of edges as G

- G' is a complete bipartite graph $(N(v), \{v_1, \dots, v_l\}, E')$. Hence, certainly G' does not have any triangles.

Hence, this shows that the maximum number of edges such that G is K_3 -free must be achieved by a complete bipartite graph with n vertices.

For any bipartite graph (L, R, E) where $|L| + |R| = n$ has exactly $|L||R|$ edges. The number of edges in bipartite graph is maximised at exactly $\lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} \rceil$. Thus, we have $\text{ex}(n, H) = \lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} \rceil = (\frac{1}{2} + o(1)) \cdot \binom{n}{2}$. $\square$

Theorem 3.59. (Turan's Theorem for K_r) Consider $H = K_r$, then the maximum number of edges in a K_r -free n -vertex graph is achieved by a complete $r - 1$ -partite graph.

Proof. We can prove this by inducting on r .

Base case: For $r = 3$, this is proven in Theorem 3.58.

Inductive step: Let G be a graph with n vertices that is K_{r+1} free. We will construct a r -partite graph that has at least the same number of edges as G .

Let v be a vertex with the maximal degree in G . Let $l = n - |N(v)|$ and $v_1, \dots, v_l$ be the copies of v . Consider a new graph $G' = (V', E')$ defined as the following,

$$\begin{aligned} V' &= N(v) \cup \{v_1, \dots, v_l\} \\ E' &= \{(u, w) : u, w \in N(v) \wedge (u, w) \in E\} \cup \{(u, v_i) : u \in N(v), i \in [l]\} = \{(u, v_i) : u \in N(v), i \in [l]\} \end{aligned}$$

Again, we will show the following

- The new graph G' has at least the same number of vertices as G and is K_{r+1} -free
- G' has number of edges maximised when G' is a complete r -partite graph.

For the first one, we have the following observation

- We have any edge in G must be one of the following
 - Connects some element in $N(v)$ to some element in $N(v)$
 - Connects some element in $V \setminus N(v)$ to some element in V

$$\begin{aligned}
|E| &= |e(N(v), N(v))| + |e(V \setminus N(v), V)| \\
&\leq |e(N(v), N(v))| + |V \setminus N(v)|d \\
&= |E'|
\end{aligned}$$

Hence, G' has at least the same number of edges as G

- Certainly $N(v)$ is K_r -free. Then G' is K_{r+1} -free as otherwise there must be at least two distinct v_i, v_j in this K_{r+1} where certainly $(v_i, v_j) \notin E$.

To prove the second point, G' has edge size of $|e(N(v), N(v))| + |V \setminus N(v)|d$. To maximise the number of edges, fixing arbitrary d and $N(v)$, where $N(v)$ must be K_r -free. By IH, $|e(N(v), N(v))|$ is maximised when $N(v)$ is a complete $r-1$ -partite graph. In this case, the new graph G' is a complete r -partite graph.

To compute the approximation of $\text{ex}(n, H)$ we have the number of edges in a complete r partite graph with n vertices is approximately

$$\binom{r-1}{2} \left(\frac{n}{r-1} \right)^2 = \frac{(r-1)(r-2)}{2} \cdot \frac{n^2}{(r-1)^2} = \frac{n^2}{2} \cdot \frac{r-2}{r-1} = \frac{n^2}{2} \cdot \left(1 - \frac{1}{r-1} \right) = \left(1 - \frac{1}{r-1} + o(1) \right) \binom{n}{2}$$

□

3.7.2. Turan's Theorem for other graphs.

We will look at Turan's Theorem for 4-cycles. Before this, we will introduce some useful results.

Theorem 3.60. (Jensen's Inequality) *For f being a convex function and X be a set of n numbers in the domain,*

$$\frac{\sum_{x \in X} f(x)}{n} \geq f\left(\frac{\sum_{x \in X} x}{n}\right)$$

Theorem 3.61. (Kővári-Sós-Turán) *Let $H = C_4$ be the 4-cycle*

$$\text{ex}(n, H) = \Theta(n^{\frac{3}{2}}) = \Theta\left(\frac{1}{\sqrt{n}}\right) \cdot \binom{n}{2}$$

Proof. We first show $\text{ex}(n, H) \in \Omega(n^{\frac{3}{2}})$ by showing that there is a graph with $\Omega(n^{\frac{3}{2}})$ edges with no 4-cycles. Any 4-cycle has 2 vertices that have 2 common neighbours. So, we want every 2 vertices to have at most 1 common neighbour. The construction requires considering points and lines in the finite field $\mathbb{F}_p$.

Take $n = 2p^2$ for some prime p . Consider $\mathbb{F}_p^2$, for arbitrary a, b define a line (with slope a and intercept b) $L_{a,b}$ as follows

$$L_{a,b} = \{(x, y) : y = ax + b\} \subseteq \mathbb{F}_p^2$$

Certainly, we have the following properties

1. **Any two distinct lines intersect at most one point:** Consider two lines $L_{a,b}, L_{a',b'}$ with $(a, b) \neq (a', b')$. Suppose (x, y) is an intersection of these two lines. Then we have

$$\begin{cases} y = ax + b \\ y = a'x + b' \end{cases}$$

This gives

$$ax + b = a'x + b' \implies (a - a')x = b' - b$$

Case 1: If $a \neq a'$, there is only one value of $x = \frac{b'-b}{a-a'}$, and hence only one corresponding value of y . This gives us exactly one intersection point.

Case 2: If $a = a'$, but $b \neq b'$. Then the above equation becomes $0 = b' - b$ which is impossible. This shows there is no such intersection point (x, y) .

In both cases, the lines intersect at most at one point.

2. **Any two distinct points with different x -values share at most one line:** Consider two points $(x, y), (x', y')$ with $x \neq x'$. Suppose $L_{a,b}$ is a common line of these two points. Then we have

$$\begin{cases} y' = ax' + b \\ y = ax + b \end{cases}$$

This gives us

$$y' - y = a(x' - x) \implies a = \frac{y' - y}{x' - x}$$

There is only one value of a , and hence only one corresponding value of b . This gives us exactly one common line.

Consider the bipartite graph (L, R, E) where $L = \mathbb{F}_p^2 = R$ where L represents the points in $\mathbb{F}_p^2$, R represents the lines in $\mathbb{F}_p^2$. Thus, we define

$$E = \{((x, y), (a, b)) \in L \times R : (x, y) \in L_{a,b}\}$$

Specifically, we have the number of vertices in this graph is $2p^2 = n$. Moreover, there are p^2 lines and each line has p points. Hence, the number of edges in this graph is $p^2 \cdot p = p^3 = \left(\frac{n}{2}\right)^{\frac{3}{2}} \in \Omega\left(n^{\frac{3}{2}}\right)$.

Finally, we show this graph does not have a 4-cycle. Specifically, we show any two vertices have at most 1 common neighbour. Take any two lines from R , then by property 1 proven previously, these two vertices have at most one common neighbour. Take any two vertices from L , then by property 2 proven previously, these two vertices have at most one common neighbour. Take one vertex from each of the L, R , there are no neighbours as there are no edges between points/lines.

We then show $\text{ex}(n, H) \in \mathcal{O}\left(n^{\frac{3}{2}}\right)$ by showing that any graph with n vertices and $5n^{\frac{3}{2}}$ edges has a 4-cycle.

First, we look at the average size of neighbours for vertices. We have

$$\sum_{v \in V} |N(v)| = 2|E| \geq 10n^{\frac{3}{2}} \implies \sum_{v \in V} \frac{|N(v)|}{n} \geq 10\sqrt{n}$$

Now, we just need to show that, for distinct $b, c \in V$, $|N(b) \cap N(c)| > 1$. Consider

$$M = |\{(a, \{b, c\}) : a, b, c \in V \wedge a \text{ is adjacent to } b, c\}|$$

This represents all combinations of, the pairs of vertices $\{b, c\}$ and one of their common neighbour a . We aim to show this is large such that some pair have more than one neighbour.

Let $d_a = |N(a)|$, $d_{b,c} = |N(b) \cap N(c)|$, using two ways of computing M (one using a , one using $\{b, c\}$), we have

$$M = \sum_{a \in V} \binom{d_a}{2} = \sum_{\{b, c\} \in \binom{V}{2}} d_{b,c}$$

We now aim to show that M is large enough so that there is some $b, c \in V$ such that $d_{b,c} > 1$. We have

$$|E| = \frac{1}{2} \sum_{a \in V} d_a \implies \sum_{a \in V} d_a \geq 10n^{\frac{3}{2}}$$

Therefore, we can use this bound on get

$$\begin{aligned}
M &= \sum_{a \in V} \binom{d_a}{2} \\
&= n \cdot \sum_{a \in V} \frac{\binom{d_a}{2}}{n} \\
&\geq n \cdot \left(\frac{\sum_{a \in V} d_a}{n} \right) \left[\binom{d_a}{2} = d_a \cdot \frac{d_a - 1}{2} \text{ is a convex function, hence Jensen's Inequality applies} \right] \\
&\geq n \cdot \binom{10\sqrt{n}}{2} \\
&= \frac{n(10\sqrt{n})(10\sqrt{n} - 1)}{2} \\
&\geq 5n^2
\end{aligned}$$

Hence, we get

$$\sum_{\{b,c\} \in \binom{V}{2}} d_{b,c} \geq 5n^2 \implies \exists d_{b,c} \geq \frac{5n^2}{\binom{n}{2}} \geq 10 > 1$$

This shows there are two vertices b, c with more than 1 neighbour, showing the existence of a 4-cycle. Thus, this gives that $\text{ex}(n, H) = \Theta(n^{\frac{3}{2}})$.

Note: $|E| \geq 5n^{\frac{3}{2}}$ gives us $d_{b,c} \geq 10$ for some $b, c \in V$. We are still able to restrict this bound by a fact (in fact, any $|E| > \frac{1}{2}n^{\frac{3}{2}}$ works). $\square$

Remark 3.62. It makes sense that this method does not generalises to arbitrary cycle C_m .

However, we can consider $C_4 = K_{2,2}$, and this method generalises to arbitrary $K_{r,s}$.

3.7.3. Random Graphs.

We modify our problem by a bit: we will fix the number of edges, and find how many 4-cycles are there.

Theorem 3.63. Suppose $G = (V, E)$ is a graph with n vertices, $p\binom{n}{2}$ edges. Then G has at least $(p^4 - o(1)) \cdot n^4$ number of 4-cycles.

Proof. Fix a graph $G = (V, E)$, let $x, y \in V$ be distinct uniformly random vertices sampled independently and denote the random variable $E_{x,y}$ as

$$R_{x,y} = \begin{cases} 1 & (x, y) \in E \\ 0 & \text{otherwise} \end{cases}$$

By construction, we have, as it is possible that $x = y$,

$$\begin{aligned}
E_x[E_y[R_{x,y}]] &= E_{x,y}[R_{x,y}] = \Pr[R_{x,y} = 1] = \left(1 - \frac{1}{n}\right) \cdot p = p - \frac{p}{n} \\
\implies E_x[(E_y[R_{x,y}])^2] &\geq (E_x[E_y[R_{x,y}]])^2 = \left(p - \frac{p}{n}\right)^2 \quad [\text{By convexity, } E[Z^2] \geq (E[Z])^2] \\
\implies E_x[E_{y_1, y_2}[R_{x, y_1} R_{x, y_2}]] &\geq \left(p - \frac{p}{n}\right)^2 \quad [\text{Taking expectation is just taking a summation}] \\
\implies E_{y_1, y_2}[E_x[R_{x, y_1} R_{x, y_2}]] &\geq \left(p - \frac{p}{n}\right)^2 \\
\implies E_{y_1, y_2}[(E_x[R_{x, y_1} R_{x, y_2}])^2] &\geq (E_{y_1, y_2}[E_x[R_{x, y_1} R_{x, y_2}]])^2 \geq \left(\left(p - \frac{p}{n}\right)^2\right)^2 \quad [\text{By convexity, } E[Z^2] \geq (E[Z])^2] \\
\implies E_{y_1, y_2}[E_{x_1, x_2}[R_{x_1, y_1} R_{x_1, y_2} R_{x_2, y_1} R_{x_2, y_2}]] &\geq \left(p - \frac{p}{n}\right)^4 \quad [\text{Taking expectation is just taking a summation}]
\end{aligned}$$

This gives us the probability when $x_1, x_2, y_1, y_2 \in V$ are sampled independently at random

$$\Pr[R_{x_1, y_1} R_{x_1, y_2} R_{x_2, y_1} R_{x_2, y_2}] \geq \left(p - \frac{p}{n}\right)^4 = p^4 - o(1)$$

Now, we have

$$\begin{aligned} \# \text{ 4-cycles in } G &= |\{\{x_1, x_2, y_1, y_2\} \subseteq V^4 : x_1, x_2, y_1, y_2 \text{ distinct}\}| \cdot \Pr[R_{x_1, y_1} R_{x_1, y_2} R_{x_2, y_1} R_{x_2, y_2}] \\ &= (n^4 - |\{\{x_1, x_2, y_1, y_2\} \subseteq V^4 : \text{at least two of the } x_1, x_2, y_1, y_2 \text{ are the same}\}|) \cdot (p^4 - o(1)) \\ &= (n^4 - \mathcal{O}(n^3)) \cdot (p^4 - o(1)) \\ &\geq n^4 \cdot (p^4 - o(1)) \end{aligned}$$

where

$$|\{\{x_1, x_2, y_1, y_2\} \subseteq V^4 : \text{at least two of the } x_1, x_2, y_1, y_2 \text{ are the same}\}| = n^4 - n(n-1)(n-2)(n-3) \in \mathcal{O}(n^3)$$

□

Remark 3.64. The above result shows that if we want to obtain a graph with edge density p with the smallest number of 4-cycles. We can take a random graph $G(n, p)$. This on expectation, gives us

$$E[\#4\text{-cycles in } G(n, p)] = (p^4 + o(1)) \cdot n^4$$

This shows that most graph are extremal (obtains minimal number of 4-cycles), and most graphs are random.

3.8. Hitting Sets.

Definition 3.65. Consider $[n]$, let $\mathcal{F}$ be a family of subsets of $[n]$, where each $F \in \mathcal{F}$ satisfies $|F| \geq \varepsilon n$ and $|\mathcal{F}| = M$. Then $S \subseteq [n]$ is called a **hitting set** of $\mathcal{F}$ if for any $F \in \mathcal{F}$, $S \cap F \neq \emptyset$.

It is obvious that we can take S as large as possible to qualify to be a hitting set. However, finding S as small as possible is much more interesting.

Theorem 3.66. *There exists a set $S \subseteq [n]$ with $|S| = \mathcal{O}\left(\frac{\log(M)}{\varepsilon}\right)$ that is a hitting set of $\mathcal{F}$.*

Proof. Let $S = \{x_1, \dots, x_t\}$ be picked from $[n]$ uniformly and independently. We want to find out the exact value of t such that $\Pr[\exists F \in \mathcal{F} : x_i \notin F, \forall 1 \leq i \leq t] < 1$ so that there is a positive probability that S is a hitting set.

Fix some $F \in \mathcal{F}$, we have

$$\Pr_{x_1, \dots, x_t} [\exists i. x_i \in F] = \prod_{i=1}^t \Pr[x_i \notin F] \leq (1 - \varepsilon)^t$$

By a union bound,

$$\Pr[\exists F \in \mathcal{F} : x_i \notin F, \forall 1 \leq i \leq t] \leq \sum_{F \in \mathcal{F}} \Pr[x_i \notin F, \forall 1 \leq i \leq t] \leq M(1 - \varepsilon)^t \leq Me^{-\varepsilon t}$$

Therefore, by setting $Me^{-\varepsilon t} < 1$, we get

$$e^{-\varepsilon t} < \frac{1}{M} \implies -\varepsilon t < -\log(M) \implies t > \frac{\log(M)}{\varepsilon}$$

Showing that by setting $t = \mathcal{O}\left(\frac{\log(M)}{\varepsilon}\right)$ we get $\Pr[\exists F \in \mathcal{F} : x_i \notin F, \forall 1 \leq i \leq t] < 1$. □

Example 3.67. Consider a $\sqrt{n} \times \sqrt{n}$ integer grid (which has n points in total). Let

$$\mathcal{F} = \{\text{all subsets of the grid of size } \geq \varepsilon n \text{ that can be contained in a circle}\}$$

We have $|\mathcal{F}| \leq n^2$ (there are n points to choose from for the center of the circle, and there are another $\leq n$ points to choose for a point on the circumference to determine the radius). Therefore, applying the above result we get

$$\frac{\log(M)}{\varepsilon} \leq \frac{\log(n^2)}{\varepsilon} = \frac{2 \log(n)}{\varepsilon}$$

This shows that there is a hitting set of size $\mathcal{O}\left(\frac{\log(n)}{\varepsilon}\right)$ for $\mathcal{F}$.

4. LINEAR ALGEBRA METHODS

4.1. Fisher's Theorem.

We first consider the following problem

Question 4.1. Let distinct $S_1, \dots, S_m \subseteq [n]$ be such that for some λ , $|S_i \cap S_j| = \lambda$ for all $i \neq j$. Then, how large can m be

Example 4.2. Here are some trivial examples, we can let

- $S_1 = \{1, 2\}, S_2 = \{1, 3\}, \dots, S_{n-1} = \{1, n\}$. This gives us $n - 1$ sets.
- $S_1 = \emptyset, S_2 = \{1\}, \dots, S_n = \{n\}$. This gives us $n + 1$ sets.

This generalises that for any collection of sets of the form $S_1 = A \cup B_1, S_2 = A \cup B_2, \dots, S_m = A \cup B_m$ where $[n] = A \sqcup B_1 \sqcup \dots \sqcup B_m$ works.

Now, here are some non-trivial examples.

- Consider $\mathbb{F}_2^r$ where $|\mathbb{F}_2^r| = 2^r = n$, for each $x \in \mathbb{F}_2^r \setminus \{0\}$ consider the set

$$S_x = \{y \in \mathbb{F}_2^r : \langle x, y \rangle = 0\}$$

Note that this has a cardinality of $2^{r-1} = \frac{n}{2}$ (as S_x has dimension $r - 1$). Similarly, we have for two distinct sets $S_x, S_{x'}, |S_x \cap S_{x'}| = 2^{r-2} = \frac{n}{4}$. Therefore, $\{S_x : x \in \mathbb{F}_2^r \setminus \{0\}\}$ gives $n - 1$ sets with pairwise intersection of size being $\frac{n}{4}$.

- (The projective plane over $\mathbb{F}_p$) Consider $\mathbb{F}_p^2$ such that $n = p^2 + p + 1$.

We consider the projective plane of $\mathbb{F}_p^2$, denoted as $\mathbb{PF}_p^2$, where any two lines intersect in 1 point (including lines with the same slope). That is, for each possible slopes of a line $m \in \mathbb{F}_p \cup \{\infty\}$, we add a point Q_m and make all lines of slope m hit at Q_m . This gives us adding a total of $p + 1$ more points.

Finally, we add one more line

$$\{Q_m : m \in \mathbb{F}_p \cup \{\infty\}\}$$

This gives us a total of $p^2 + p + 1$ points in $\mathbb{PF}_p^2$ (p^2 points in $\mathbb{F}_p^2$ and $p + 1$ newly constructed Q_m).

Similarly, there are $p^2 + p + 1$ lines in $\mathbb{PF}_p^2$ ($p^2 + p$ lines of the form $mx + b$ and 1 newly constructed line).

We can also observe that

1. For every point there are exactly $p + 1$ lines that touches the point
2. For any line, there are exactly $p + 1$ points on it
3. Any two points determines a line
4. Any two lines intersect at a point

Hence, consider the set $S_1, \dots, S_{p^2+p+1}$ to be the lines of $\mathbb{PF}_p^2$. This gives us $p^2 + p + 1 = n$ sets of pairwise intersection of size 1.

Theorem 4.3. (Fishers Theorem) *Let distinct $S_1, \dots, S_m \subseteq [n]$ be such that $|S_i \cap S_j| = \lambda$ for all $i \neq j$. Then, $m \leq n + 1$. Furthermore, if all S_i 's are nonempty then $m \leq n$.*

Proof. Take any sets $S_1, \dots, S_m \subseteq [n]$ that satisfies the requirements. Let $v_i \in \{0, 1\}^n \subseteq \mathbb{R}^n$ be the indicator vector of S_i . Therefore, we get

$$\langle v_i, v_j \rangle = \lambda$$

for all $i \neq j$. Consider the matrix $M \in \mathbb{R}^{n \times m}$ defined as

$$M = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_m \\ | & | & \cdots & | \end{pmatrix}$$

then we get $M^\top M \in \mathbb{R}^{m \times m}$ defined as

$$M^\top M = \begin{pmatrix} a_1 & \lambda & \lambda & \cdots & \lambda \\ \lambda & a_2 & \lambda & \cdots & \lambda \\ \lambda & \lambda & a_3 & \cdots & \lambda \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda & \lambda & \lambda & \cdots & a_m \end{pmatrix}$$

where for all $a_i = \langle v_i, v_i \rangle = |S_i|$. Since $M \in \mathbb{R}^{n \times m}$ we have $\text{rank}(M^\top M) \leq n$. Now, we will consider several cases for $M^\top M$ based on the cardinality of S_i 's. Note that for any S_i , $|S_i| \geq \lambda$.

Case 1: Suppose all i , $a_i > \lambda$. We show $M^\top M \in \mathbb{R}^{m \times m}$ has full rank m , which immediately gives $m \leq n$. Suppose $x = (x_1, \dots, x_m)^\top \in \mathbb{R}^m$ satisfies

$$(M^\top M)x = 0$$

We show this forces $x = 0$. Let $X = \sum_{i=1}^m x_i$. The i -th coordinate of $(M^\top M)x$ gives

$$0 = ((M^\top M)x)_i = \sum_{j \in [m]} (M^\top M)_{ij} x_j = a_i x_i + \lambda \sum_{j \neq i} x_j = a_i x_i + \lambda(X - x_i)$$

Since $a_i - \lambda > 0$, we get

$$0 = (a_i - \lambda)x_i + \lambda X \implies x_i = -\frac{\lambda X}{a_i - \lambda}$$

Therefore,

$$X = \sum_{i \in [m]} x_i = -\lambda X \sum_{i \in [m]} \frac{1}{a_i - \lambda} \implies X \left(1 + \lambda \sum_{i \in [m]} \frac{1}{a_i - \lambda} \right) = 0$$

Because $a_i - \lambda > 0$ and $\lambda \geq 0$, the term in the parentheses is strictly positive. Hence, we have $X = 0$. Substituting $X = 0$ back into our equation for x_i yields $x_i = 0$ for all i . Thus, $x = 0$, proving $M^\top M$ has full rank m .

Case 2: Suppose there is an S_i where $|S_i| = \lambda$. In this case, we cannot consider the matrix just like case 1. WLOG, assuming $|S_1| = \lambda$. Then for all j , $|S_1 \cap S_j| = \lambda \implies S_1 \subseteq S_j$.

Therefore, we can write

$$S_1 = S_1, S_2 = S_1 \cup (S_2 \setminus S_1), \dots, S_m = S_1 \cup (S_m \setminus S_1)$$

note that we must have $S_2 \setminus S_1, \dots, S_m \setminus S_1 \neq \emptyset$. We also have for any $S_j, S_{j'}$

$$|S_j \cap S_{j'}| = \lambda \implies (S_j \setminus S_1) \cap (S_{j'} \setminus S_1) = \emptyset$$

Therefore, this gives

$$S_1 \sqcup (S_2 \setminus S_1) \sqcup \dots \sqcup (S_m \setminus S_1) \subseteq [n]$$

To argue the bound of m in cases:

- If there is an empty set in S_i 's, then only S_1 can be empty, and $m \leq n + 1$.
- If there is no empty sets, then $m \leq n$.

□

Remark 4.4. The bound obtained in Fisher's Theorem is strict.

4.2. Odd Town / Even Town.

We will consider the following problem

Question 4.5. (Odd-Even Town) Consider a town with n people and m clubs $C_1, \dots, C_m$.

- No two clubs have the same membership
- All clubs must have an odd number of members
- Any two distinct clubs must have an even number of common members

How large can m be

Example 4.6. Here are some examples:

- We can let $C_1 = \{1, 2, 3\}, C_2 = \{1, 2, 4\}, \dots, C_{n-2} = \{1, 2, n\}$. This gives us $n - 2$ sets.
- We can let $C_1 = \{1\}, \dots, C_n = \{n\}$. This gives us n sets.

Theorem 4.7. (Odd-Even Town) *The maximum number of clubs we can have is n .*

Proof. Under a membership assignment that satisfies the conditions, let $v_i \in \mathbb{F}_2^n$ be the indicator vector for the club C_i , we have

- $\forall i \neq j, \langle v_i, v_j \rangle = 0$
- $\forall i, \langle v_i, v_i \rangle = 1$

Consider the matrix $M \in \mathbb{F}_2^{n \times m}$ defined as

$$M = \begin{pmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_m \\ | & | & \dots & | \end{pmatrix}$$

In this case $M^\top M = I$, which has rank m . On the other hand, rank must be at most n . Hence, $m \leq n$.

Example 4.6 showed that m can be as large as n . □

Now, we consider a variant

Question 4.8. (Even-Even Town) Consider a town with n people and m clubs $C_1, \dots, C_m$.

- No two clubs have the same membership
- All clubs must have an even number of members
- Any two distinct clubs must have an even number of common members

How large can m be

Example 4.9. Here is an example:

- We can partition $[n]$ into $\lfloor \frac{n}{2} \rfloor$ pairs.

$$\{1, 2\}, \{3, 4\}, \dots, \left\{2\left\lfloor \frac{n}{2} \right\rfloor - 1, 2\left\lfloor \frac{n}{2} \right\rfloor\right\}$$

This gives us $2^{\lfloor \frac{n}{2} \rfloor}$ sets.

Theorem 4.10. *The maximum number of clubs we can have is $m \leq 2^{\lfloor \frac{n}{2} \rfloor}$*

Proof. Under a membership assignment that satisfies the conditions, let $v_i \in \mathbb{F}_2^n$ be the indicator vector for the club C_i , we have

- $\forall i \neq j, \langle v_i, v_j \rangle = 0$
- $\forall i, \langle v_i, v_i \rangle = 0$

In this case $M^\top M = 0$. So we cannot tell anything about the value of m . Instead, we will use a different approach.

Let $S = \{v_1, \dots, v_m\}$. Let $\bar{S} = \text{span}(S)$ over the $\mathbb{F}_2^n$ space. We show that $\bar{S} \perp \bar{S}$ ($\bar{S}$ is self-orthogonal). Indeed, for any $\sum_{i=1}^m a_i v_i, \sum_{i=1}^m b_i v_i \in \bar{S}$ we have

$$\left\langle \sum_{i=1}^m a_i v_i, \sum_{i=1}^m b_i v_i \right\rangle = \sum_{i=1}^m \sum_{j=1}^m a_i b_j \langle v_i, v_j \rangle = 0$$

Consider $\bar{S}^\perp$ (the space that is orthogonal to $\bar{S}$), we have the identity $\dim(\bar{S}) + \dim(\bar{S}^\perp) = n$. Since we showed that $\bar{S} \subseteq \bar{S}^\perp$, $n - \dim(\bar{S}) \geq \dim(\bar{S}) \implies \dim(\bar{S}) \leq \lfloor \frac{n}{2} \rfloor$. Therefore,

$$|S| \leq |\bar{S}| \leq 2^{\dim(\bar{S})} \leq 2^{\lfloor \frac{n}{2} \rfloor}$$

□

Again, we consider a variant

Question 4.11. (Even-Odd Town) Consider a town with n people and m clubs $C_1, \dots, C_m$.

- No two clubs have the same membership
- All clubs must have an even number of members
- Any two distinct clubs must have an odd number of common members

How large can m be

Theorem 4.12. (Even-Odd Town) *The maximum number of clubs we can have is $m \leq n$.*

Proof. An example of club membership $C_1, \dots, C_m$ is that

- When n is odd,

$$C_i = \{1, \dots, n\} \setminus \{i\}, \forall i = 1, \dots, n$$

This gives n clubs and for all $i = 1, \dots, n$, $|C_i|$ is even. For all i, j , $|C_i \cap C_j|$ is odd.

- When n is even,

$$C_i = \{1, \dots, n-1\} \setminus \{i\}, \forall i = 1, \dots, n-1, C_n = \{n\}$$

This gives n clubs and for all $i = 1, \dots, n$, $|C_i|$ is even. For all i, j , $|C_i \cap C_j|$ is odd.

To prove the bound, consider a membership assignment that satisfies the conditions, let $v_i \in \mathbb{F}_2^n$ be the indicator vector for the club C_i , we have

- $\forall i \neq j, \langle v_i, v_j \rangle = 1$
- $\forall i, \langle v_i, v_i \rangle = 0$

Consider the matrix $M \in \mathbb{F}_2^{n \times m}$ defined as

$$M = \begin{pmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_m \\ | & | & \dots & | \end{pmatrix}$$

In this case $M^\top M = \mathbb{1} - I$.

1. If m is even, $m-1$ is odd. Summing all rows produces the all-ones vector, so the rows span $\mathbb{F}_m^2$ and $\text{rank}(M^\top M) = m$. Thus $m \leq n$.
2. If m is odd, $m-1$ is even. The sum of all rows is the zero vector, so $\text{rank}(M^\top M) \leq m-1$, yielding $m \leq n+1$.

We first do that case that n is odd. Suppose there is a membership assignment with $m = n+1$ clubs (which is even), by the first property we obtain $m \leq n$, a contradiction. Hence, when n is odd, $m \leq n$.

Now, suppose n is even. Certainly, both property implies that $m \leq n+1$ (**this is a weaker bound**). □

The final variant we are considering is

Question 4.13. (Odd-Odd Town) Consider a town with n people and m clubs $C_1, \dots, C_m$.

- No two clubs have the same membership
- All clubs must have an odd number of members
- Any two distinct clubs must have an odd number of common members

How large can m be

Theorem 4.14. (Odd-Odd Town) $m \leq 2^{\lfloor \frac{n-1}{2} \rfloor} + 1$.

Proof. Under a membership assignment that satisfies the conditions, let $v_i \in \mathbb{F}_2^n$ be the indicator vector for the club C_i , we have

- $\forall i \neq j, \langle v_i, v_j \rangle = 1$
- $\forall i, \langle v_i, v_i \rangle = 1$

In this case $M^\top M = 1$. So we cannot tell anything about the value of m . Instead, we will use a different approach.

Let $S = \{v_1, \dots, v_m\}$. Let $S' = \{w_1 = v_2 - v_1, w_2 = v_3 - v_1, \dots, w_{m-1} = v_m - v_1\}$. Note that S' satisfies

- $\forall i \neq j, \langle w_i, w_j \rangle = 0$
- $\forall i, \langle w_i, w_i \rangle = 0$

Showing that $S' \subseteq (S')^\perp$. Moreover, since for all $w_i, \langle v_1, w_i \rangle = \langle v_1, v_i \rangle - \langle v_1, v_1 \rangle = 0$. This shows that S' lie in an $n - 1$ dimensional space. Therefore, by **Theorem 4.10**, we have

$$|S| = |S'| + 1 \leq |\overline{S'}| + 1 \leq 2^{\dim(\overline{S'})} + 1 \leq 2^{\lfloor \frac{n-1}{2} \rfloor} + 1$$

□

4.3. 1-Distance Sets in $\mathbb{R}^n$.

In this section, we are interested in answering the following

Question 4.15. Given $x_1, \dots, x_m \in \mathbb{R}^n$ with $\|x_i - x_j\|_2$ all equal, how large can m be

Example 4.16.

- In $\mathbb{R}^2, m \leq 3$: given any 2 points A, B , the circle centred at A with radius AB and the circle centred at B with radius AB intersects at most 2 points. And we can only choose one of them.
- In $\mathbb{R}^3, m \leq 4$. Given any 3 points A, B, C with equal distance, the sphere centred at A with radius AB , the circle centred at B with radius AB and the circle centred at C with radius AB intersects at most 2 points. And we can only choose one of them.

It seems like the general idea is the obtain n points within $n - 1$ dimensional plane that is equal in distance, then we obtain one and only one final point in the remaining dimension while making sure the distance is still equal. This gives us the following theorem

Theorem 4.17. For any 1-distance set of $\mathbb{R}^n$, the size m is at most $m \leq n + 1$

Proof. We first give a collection of $n + 1$ points, that all distances are equal. Consider for all $1 \leq i \leq n$

$$v_i = e_i$$

Certainly for all $1 \leq i, j \leq n$ where $i \neq j$, we have

$$\|v_i - v_j\| = \sqrt{2}$$

We now find the last point v_{n+1} such that for all $1 \leq i \leq n, \|v_i - v_{n+1}\| = \sqrt{n}$.

Note that v_1, v_{n+1} are on the plane $\sum_{i=1}^n x_i = 1$, points that is perpendicular to such plane has the form $(x, x, \dots, x)$. Now, we find such value of x which has exactly $\sqrt{2}$ in distance to $v_1, \dots, v_n$. We get for any v_i

$$\begin{aligned} \|(x, x, \dots, x) - v_i\| &= \sqrt{(x-1)^2 + (n-1)x^2} = \sqrt{2} \\ \implies (x-1)^2 + (n-1)x^2 &= 2 \\ \implies nx^2 - 2x - 1 &= 0 \\ \implies x &= \frac{2 \pm \sqrt{4+4n}}{2n} = \frac{1 \pm \sqrt{1+n}}{n} \end{aligned}$$

Hence, we can take $v_{n+1} = \frac{1+\sqrt{4+4n}}{4}$ and this gives us the collection of $n+1$ points.

Suppose $x_1, \dots, x_{n+2} \in \mathbb{R}^n$ such that all distances are equal, WLOG by scaling, we can assume that for all $i \neq j$

$$\|x_i - x_j\| = 1$$

Define $v_1, \dots, v_{n+1}$ with

$$v_i = x_i - x_{n+2}$$

We get

1. For all i , $\|v_i\| = 1$
2. For all $i \neq j$, $\|v_i - v_j\| = \|x_i - x_j\| = 1$

Then, for all $i \neq j$

$$1 = \|v_i - v_j\|^2 = \langle v_i - v_j, v_i - v_j \rangle = \langle v_i, v_i \rangle + \langle v_j, v_j \rangle - 2\langle v_i, v_j \rangle = 2 - 2\langle v_i, v_j \rangle \implies \langle v_i, v_j \rangle = \frac{1}{2}$$

Similarly, for all i , $\langle v_i, v_i \rangle = \|v_i\|^2 = 1$. Therefore, consider the following matrix $M \in \mathbb{R}^{n \times (n+1)}$

$$M = \begin{pmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_{n+1} \\ | & | & \dots & | \end{pmatrix}$$

We get

$$M^\top M = (\langle v_i, v_j \rangle)_{i,j=1}^{n+1} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \dots & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} & \dots & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 & \dots & \frac{1}{2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \dots & 1 \end{pmatrix}$$

We now show this matrix is positive definite and hence has full rank. For any non-zero $x \in \mathbb{R}^{n+1}$ we get

$$x^\top M^\top M x = x^\top \left(\frac{1}{2}(\mathbb{1} + \mathbb{I}) \right) x = \frac{1}{2}x^\top \mathbb{1}x + \frac{1}{2}x^\top \mathbb{I}x = \frac{1}{2} \left(\sum_{i=1}^{n+1} x_i \right)^2 + \frac{1}{2} \left(\sum_{i=1}^{n+1} x_i^2 \right) \geq \frac{1}{2} \left(\sum_{i=1}^{n+1} x_i^2 \right) > 0$$

However, since $M \in \mathbb{R}^{n \times (n+1)}$ where $\text{rank}(M) \leq n$ as the vector $v_1, \dots, v_{n+1}$ lies in $\mathbb{R}^n$, we have $\text{rank}(M^\top M) \leq n$, a contradiction. $\square$

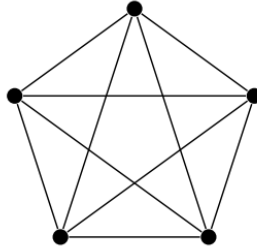
4.4. 2-Distance Sets of $\mathbb{R}^n$.

In this section, we are interested in answering the following

Question 4.18. Given $x_1, \dots, x_m \in \mathbb{R}^n$ with $|\{\|x_i - x_j\| : i \neq j\}| \leq 2$, how large can m be

Example 4.19.

- In $\mathbb{R}^2$, m can be 5 under the following configuration



- Consider the set

$$\left\{ e_{ij} = e_i + e_j : \{i, j\} \in \binom{[n]}{2} \right\}$$

This is a 2-distance set, for distinct pairs $\{i, j\}, \{i', j'\}$ we have

$$\|e_{ij} - e_{i'j'}\| = \begin{cases} 2 & \text{if } |\{i, j\} \cap \{i', j'\}| = 0 \\ \sqrt{2} & \text{if } |\{i, j\} \cap \{i', j'\}| = 1 \end{cases}$$

only the two distances can be possible. Therefore this is a 2-distance set of size

$$m = \binom{n}{2} = \Omega(n^2)$$

Theorem 4.20. *For any 2-distance set of $\mathbb{R}^n$, the size m is at most $m \leq \mathcal{O}(n^2)$*

Proof. Define a function P as a polynomial computing the square of the distance

$$P(\vec{X}, \vec{Y}) = P(x_1, \dots, x_n, y_1, \dots, y_n) = \sum_{i=1}^n (x_i - y_i)^2$$

Therefore, for any fixed $\vec{y} \in \mathbb{R}^n$, we can define polynomial

$$P_y(\vec{X}) = P(\vec{X}, \vec{y})$$

Suppose $v_1, \dots, v_m$ have the property that $\forall i \neq j, \|v_i - v_j\| \in \{A, B\}$ for some non-zero A, B .

Define for all $i = 1, \dots, m$, $H_i(\vec{X}) = (P_{v_i}(\vec{X}) - A^2)(P_{v_i}(\vec{X}) - B^2)$. Therefore, we have

$$H_i(v_j) = \begin{cases} A^2 B^2 \neq 0 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

The idea is that for all $i = 1, \dots, m$, H_i is a polynomial of some variables. We can treat this as some vector space and aim to show that

1. H_i 's are linearly independent
2. By 1, the number of H_i 's will be restricted to the dimension of the vector space spanned by H_i 's

We first find the dimension of such vector space. Expanding the expression of $H_i(\vec{X})$, we have

$$\begin{aligned} H_i(\vec{X}) &= (P_{v_i}(\vec{X}) - A^2)(P_{v_i}(\vec{X}) - B^2) \\ &= \left(\sum_{j=1}^n (X_j - v_{ij})^2 - A^2 \right) \left(\sum_{j=1}^n (X_j - v_{ij})^2 - B^2 \right) \\ &= \left(\sum_{j=1}^n X_j^2 - 2 \sum_{j=1}^n X_j v_{ij} + \left(\sum_{j=1}^n v_{ij}^2 - A^2 \right) \right) \left(\sum_{j=1}^n X_j^2 - 2 \sum_{j=1}^n X_j v_{ij} + \left(\sum_{j=1}^n v_{ij}^2 - B^2 \right) \right) \\ &= \left(\sum_{j=1}^n X_j^2 \right)^2 + \sum_{j=1}^n c_j \left(\sum_{k=1}^n X_k^2 \right) X_j + \sum_{1 \leq j, k \leq n} d_{j,k} X_j X_k + \sum_{j=1}^n e_j X_j + f \end{aligned}$$

for some constants $c_j, d_{j,k}, e_j, f \in \mathbb{R}$.

This shows that all the $H_1, \dots, H_m$ lie in the $\mathbb{R}$ -span of the following polynomials

- $\left(\sum_{j=1}^n X_j^2 \right)^2$
- $\left(\sum_{j=1}^n X_j^2 \right) X_k$ for $k \in [n]$
- $X_j X_k$ for $\{j, k\} \in \binom{[n]}{2}$
- X_j^2 for $j \in [n]$
- X_j for $j \in [n]$
- 1

Hence, we get

$\dim(\text{span}(H_1, \dots, H_m)) \leq \# \text{ of elements in the above list} = 1 + n + \binom{n}{2} + n + n + 1 = \binom{n}{2} + 3n + 2$

Now, we just need to show that $H_1, \dots, H_m$ are linearly independent. Suppose $\alpha_1, \dots, \alpha_m$ satisfies

$$\sum_{i=1}^n \alpha_i H_i(\vec{X}) = 0$$

Then, since $H_i(v_j) = 0$ for all $i \neq j$, we have for any $1 \leq j \leq m$

$$\sum_{i=1}^n \alpha_i H_i(v_j) = 0 \implies \alpha_j H_j(v_j) = 0 \implies \alpha_j A^2 B^2 = 0 \implies \alpha_j = 0$$

Thus, $H_1, \dots, H_m$ are linearly independent, and we get $m \leq \binom{n}{2} + 3n + 2 \in \mathcal{O}(n^2)$. $\square$

Remark 4.21. This polynomial-method argument is part of a broader ideas related to s -distance sets.

Blokhuis developed analogous methods for bounding the size of s -distance sets in Euclidean space.

4.5. VC-Dimension.

4.5.1. Range Spaces, VC-Dimension and Sauer-Shelah Lemma.

Just some information the notion of VC dimension (founded by Vapnik and Chervonenkis) comes from learning theory.

Definition 4.22. A **range space** $(X, \mathcal{R})$ has elements

- X called a universe
- $\mathcal{R} \subseteq \mathcal{P}(X)$ called a range space which is a family of subsets of X

Example 4.23. (Half Spaces) Let $X = \mathbb{R}^n$, we can define $\mathcal{R} = \{H_{a,b} : a, b \in \mathbb{R}\}$ which is a family of half spaces.

$$H_{a,b} = \{x \in \mathbb{R}^n : \langle a, x \rangle \geq b\}$$

Then, we are interested in answering the following.

Question 4.24. Given some hidden set $R \in \mathcal{R}$. Given a distribution on X , we sample $x_1, \dots, x_n$ from the distribution and learn whether $x_i \in R$ or not.

How many samples is enough for us to predict with a fixed probability of correctness whether $y \in R$ for a new sample y .

Definition 4.25. Given a range space $(X, \mathcal{R})$, a subset $S \subseteq X$ is **shattered** by $\mathcal{R}$ if

$$\{S \cap R : R \in \mathcal{R}\} = \mathcal{P}(S)$$

Example 4.26. (Half Spaces) Consider the range space defined in Example 4.23.

- A singleton is shattered
- A set of size 2 is shattered
- A set of size 3 is not shattered
 - If $\{a, b, c\}$ are colinear where b is in the middle of a and c , we cannot have a half space that contains only b

Example 4.27. (Lines) Consider a range space $(X, \mathcal{R})$ defined as $X = \mathbb{R}^2$, $\mathcal{R} = \{\text{lines in } \mathbb{R}^2\} = \{\{ax + b : x \in \mathbb{R}\} : a, b \in \mathbb{R}\}$.

- A singleton is shattered
- A set of size 2 is shattered
- A set of size 3 is not shattered
 - If $\{a, b, c\}$ are not colinear there is no line that passes through all a, b, c .
 - If $\{a, b, c\}$ are colinear there is no line that passes through any proper subset of $\{a, b, c\}$.

Example 4.28. (More VC-dimension Examples)

- $\text{VCdim}(\mathbb{R}, \{[a, b] : a, b \in \mathbb{R}\}) = 2$: See **Example 4.35** for explanation.
- $\text{VCdim}(\mathbb{R}^2, \text{convex sets}) = \infty$: For any $n \in \mathbb{N}$, line them on a circle. For any subset of the set of vertices, take the convex hull of the selected vertices.
- $\text{VCdim}(\mathbb{R}^2, \text{disks}) = 3$? No 4-point set can be shattered using what we studied in class.

Definition 4.29. (VC-dimension) The **VC-dimension** of a range space $(X, \mathcal{R})$ is the largest d such that there exists a subset of size $d \subseteq X$ that is shattered.

Example 4.30.

1. $\text{VCdim}(\mathbb{R}^2, \text{lines}) = \text{VCdim}(\mathbb{R}^2, \{\{ax + b : x \in \mathbb{R}\} : a, b \in \mathbb{R}\}) = 2$, as there is no set of size 3 can be shattered.
2. $\text{VCdim}(\mathbb{R}^2, \text{half spaces}) = \text{VCdim}(\mathbb{R}^2, \{H_{a,b} : a, b \in \mathbb{R}\}) = 3$

It seems easy for lower dimensions, now we would like to answer the following questions.

Question 4.31.

- What is $\text{VCdim}(\mathbb{R}^n, \text{half spaces})$?
- What is $\text{VCdim}(\mathbb{R}^n, \text{hyper planes})$, where for a fixed $a, b \in \mathbb{R}$, $\{x : \langle a, x \rangle = b\}$ is a hyper plane?

Lemma 4.32. (Sauer-Shelah Lemma) *If $(X, \mathcal{R})$ has VC-dimension d , then for every $S \subseteq X$ with $|S| = n$,*

$$|\{S \cap R : R \in \mathcal{R}\}| \leq \sum_{i=0}^d \binom{n}{i} \in \mathcal{O}(n^d)$$

Proof. We first transform this lemma into a more combinatorial language.

Fix $S \subseteq X$ with $|S| = n$. For each $R \in \mathcal{R}$ define vector $v_R \in \{0, 1\}^n$ be an indicator vector of $S \cap R$. And denote $Y = \{v_R : R \in \mathcal{R}\}$. Note that $|Y| = |\{S \cap R : R \in \mathcal{R}\}|$. And therefore we just need to show that

$$|Y| \leq \sum_{i=0}^d \binom{n}{i}$$

First, note that since $(X, \mathcal{R})$ has VC-dimension d , no $d+1$ -element subset of S is shattered by $\mathcal{R}$. Thus, for any $I \subseteq [n]$ with $|I| = d+1$ there is some $z_I \in \{0, 1\}^I$ such that

$$\forall y \in Y, y|_I \neq z_I \implies \{y|_I\} \neq \{0, 1\}^I$$

Now, using this assumption we prove that $|Y| \leq \sum_{i=0}^d \binom{n}{i}$.

We will prove by contrapositive that if $|Y| > \sum_{i=0}^d \binom{n}{i}$, there is some subset $I \subseteq [n]$ with $|I| = d+1$ such that $\{y|_I : y \in Y\} = \{0, 1\}^{d+1}$. This will be proven by induction

Base case: If $d = 0$, then $|Y| > \sum_{i=0}^0 \binom{n}{i} = 1$. Then, choose $y, y' \in Y$ such that $y \neq y'$. They must differ in some coordinate i . Thus, taking $\{i\} \subseteq [n]$ where $1 = |I| = 0 + 1$ we get

$$\{y|_{\{i\}} : y \in Y\} = \{0, 1\}$$

If $n = 0$, there is no subset $I \subseteq [n]$ with $|I| = d+1 > 0$. Thus, the statement is vacuously true.

Inductive step: We will consider the sets

$$Y_0 = \{x \in \{0, 1\}^{n-1} : 0x \in Y\} \quad Y_1 = \{x \in \{0, 1\}^{n-1} : 1x \in Y\}$$

Since every element in Y either starts with 0 or 1, we get

$$|Y| = |Y_0| + |Y_1|$$

Since there are some elements that appears in both Y_0 and Y_1

$$|Y| = |Y_0 \cap Y_1| + |Y_0 \cup Y_1|$$

By assumption

$$|Y_0 \cap Y_1| + |Y_0 \cup Y_1| = |Y| > \sum_{i=0}^d \binom{n}{d} = \sum_{i=0}^{d-1} \binom{n-1}{i} + \sum_{i=0}^d \binom{n-1}{i} \quad [\text{By combinatorial identity}]$$

We now argue in cases

Case 1: If $|Y_0 \cap Y_1| > \sum_{i=0}^{d-1} \binom{n-1}{i}$. By IH, there exists a set $I' \subseteq \{2, \dots, n\}$ (we will the original index in Y) with $|I'| = (d-1) + 1 = d$ such that

$$\{x|_{I'} : x \in Y_0 \cap Y_1\} = \{0, 1\}^{I'}$$

Take $I = I' \cup \{1\}$, then $|I| = d + 1$. We show that $\{y|_I : y \in Y\} = \{0, 1\}^I$. Indeed, for any $z \in \{0, 1\}^I$, by the construction of I' , there is some $x \in Y_0 \cap Y_1$ that $x|_{I'} = z|_{I'}$. So we either have $0x|_I = z|_I$ or $1x|_I = z|_I$. Note that $0x, 1x \in Y$, we are done.

Case 2: If $|Y_0 \cap Y_1| > \sum_{i=0}^d \binom{n-1}{i}$, then by IH, there is some $I \subseteq \{2, \dots, n\}$ of size $d + 1$ such that

$$\{x|_I : x \in Y_0 \cap Y_1\} = \{0, 1\}^I$$

We show that $\{y|_I : y \in Y\} = \{0, 1\}^I$. Indeed, for any $z \in \{0, 1\}^I$, there is some $x \in Y_0 \cap Y_1$ that $x|_I = z|_I$. So we have $0x|_I = z|_I$ and $1x|_I = z|_I$. Note that one of the $0x, 1x \in Y$, we are done. $\square$

4.5.2. Shatter Function.

Definition 4.33. (Shatter Function) A shatter function $f : \mathbb{N} \rightarrow \mathbb{N}$ of $(X, \mathcal{R})$ is defined as

$$f(n) = \max_{S \subseteq X, |S|=n} |\{S \cap R : R \in \mathcal{R}\}|$$

Remark 4.34. By the definition of shatter function, we get for all $(X, \mathcal{R})$ with a shatter function f ,

- $f(n) \leq 2^n$ as $\{S \cap R : R \in \mathcal{R}\} \subseteq \mathcal{P}(S)$
- $\text{VCdim}(X, \mathcal{R}) = \max\{n : f(n) = 2^n\}$

Example 4.35. Take $(X, \mathcal{R}) = (\mathbb{R}, \{[a, b] : a, b \in \mathbb{R}\})$ where range space is defined by the closed interval. We have

- $f(1) = 2$: Any singletons $S = \{a\}$ we can easily find two closed intervals where each contains a and does not contain a respectively.
- $f(2) = 4$: Any $S = \{a, b\}$ (WLOG, $a < b$) we can easily find four closed intervals where each contains a only, b only, both and neither respectively.
- $f(3) = 7$: Any $S = \{a, b, c\}$ (WLOG $a < b < c$) we can find intervals that contains any subset of S except for $\{a, c\}$.

Proposition 4.36. (Polynomial-vs-Exponential Dichotomy) For any range space $(X, \mathcal{R})$, the shatter function f satisfies exactly one of the following

1. $f(n) = 2^n$ for all $n \geq 1$. In this case, $\text{VCdim}(X, \mathcal{R}) = \infty$
2. $f(n) \leq \mathcal{O}(n^d)$ for all $n \geq 1$. In this case, $\text{VCdim}(X, \mathcal{R}) = d$, which is finite.

Proof. We prove by cases.

Case 1: If $\text{VCdim}(X, \mathcal{R}) = \infty$, then for every $n \geq 1$ there exists a subset $A \subseteq X$ with $|A| = n$ that is shattered. Hence, $f(n) \geq 2^n$. Since $f(n) \leq 2^n$, $f(n) = 2^n$.

Case 2: Now, suppose $\text{VCdim}(X, \mathcal{R}) = d < \infty$, by Sauer-Shelah Lemma, for n sufficiently large (larger than d),

$$f(n) \leq \sum_{i=0}^d \binom{n}{i} \leq \sum_{i=0}^d \frac{n^i}{i!} = n^d \cdot \sum_{i=0}^d \frac{1}{i! \cdot n^{d-i}} \leq n^d \cdot \sum_{i=0}^d \frac{1}{i!} \leq e \cdot n^d \in \mathcal{O}(n^d)$$

$\square$

4.5.3. Intersection Of Range Spaces.

Definition 4.37. (Intersection Range Space) Given two range spaces $(X, \mathcal{R}_1)$ and $(X, \mathcal{R}_2)$, the intersection range space is defined as

$$\mathcal{R} = \{R_1 \cap R_2 : R_1 \in \mathcal{R}_1, R_2 \in \mathcal{R}_2\}$$

Theorem 4.38. *Let $(X, \mathcal{R}_1)$ and $(X, \mathcal{R}_2)$ be two range spaces with*

$$\text{VCdim}(X, \mathcal{R}_1) \leq d \quad \wedge \quad \text{VCdim}(X, \mathcal{R}_2) \leq d.$$

Consider the intersection range space $\mathcal{R}$. Then, for some constant C

$$\text{VCdim}(X, \mathcal{R}) \leq Cd \log(d+1) \in \mathcal{O}(d \log(d))$$

Proof. Let f_1, f_2, f be the shatter functions of $(X, \mathcal{R}_1)$, $(X, \mathcal{R}_2)$, and $(X, \mathcal{R})$ respectively. Denote f to be the shatter function of $(X, \mathcal{F})$. We show that for all $n \in \mathbb{N}$, $f(n) \leq f_1(n)f_2(n)$.

Fix any $n \in \mathbb{N}$ take some $S \subseteq X$ with $|S| = n$. We have

$$\begin{aligned} \{S \cap R : R \in \mathcal{R}\} &= \{S \cap (R_1 \cap R_2) : R_1 \in \mathcal{R}_1 \wedge R_2 \in \mathcal{R}_2\} \\ &= \{(S \cap R_1) \cap (S \cap R_2) : R_1 \in \mathcal{R}_1 \wedge R_2 \in \mathcal{R}_2\} \end{aligned}$$

Therefore, every $S \cap R$ for some $R \in \mathcal{R}$ is determined by a pair of $S \cap R_1$ for some $R_1 \in \mathcal{R}_1$ and $S \cap R_2$ for some $R_2 \in \mathcal{R}_2$. This gives,

$$\begin{aligned} |\{S \cap R : R \in \mathcal{R}\}| &\leq |\{S \cap R_1 : R_1 \in \mathcal{R}_1\}| |\{S \cap R_2 : R_2 \in \mathcal{R}_2\}| \\ \Rightarrow \max_{\substack{S \subseteq X \\ |S|=n}} |\{S \cap R : R \in \mathcal{R}\}| &\leq \max_{\substack{S \subseteq X \\ |S|=n}} |\{S \cap R_1 : R_1 \in \mathcal{R}_1\}| \max_{\substack{S \subseteq X \\ |S|=n}} |\{S \cap R_2 : R_2 \in \mathcal{R}_2\}| \quad [\text{Take } S \text{ as arbitrary}] \\ \Rightarrow f(n) &\leq f_1(n)f_2(n) \end{aligned}$$

as required.

Now, by Sauer-Shelah Lemma, there is a fixed positive constant C such that

$$f(n) \leq f_1(n)f_2(n) \leq Cn^{\text{VCdim}(X, \mathcal{F}_1)} n^{\text{VCdim}(X, \mathcal{F}_2)} = Cn^{2d}$$

Furthermore, suppose $\text{VCdim}(X, \mathcal{F}) = m$. We have $f(m) = 2^m \leq Cm^{2d}$. Now, we just need to find the upper bound for m . We will do this by showing that taking $m \in \Omega(d \log(d))$ we have $2^m > Cm^{2d}$. Indeed, we have

$$2^m > Cm^{2d} \Rightarrow m > \log_2(C) + 2d \log_2(m)$$

By substituting $m = C'd \log_2(d)$ we have

$$\log_2(C) + 2d \log_2(m) = \log_2(C) + 2d(\log_2(d) + \log_2(C')) + \log_2 \log_2(d) \leq 2d \log_2(d) + \mathcal{O}(d \log_2 \log_2(d))$$

Thus, by taking C' to be large enough, we are able to get

$$\log_2(C) + 2d \log_2(m) \leq 2d \log_2(d) + \mathcal{O}(d \log_2 \log_2(d)) < C'd \log_2(d) = m \Rightarrow 2^m > Cm^{2d}$$

Hence, this gives us $\text{VCdim}(X, \mathcal{F})$ is at most $\mathcal{O}(d \log(d))$. □

4.5.4. Epsilon-nets.

Definition 4.39. (Epsilon-nets) Let $(X, \mathcal{R})$ be a range space of VC-dimension d , and let $S \subseteq X$ be a finite set with $|S| = n$. A subset $T \subseteq S$ is called an ε -net if for every range $R \in \mathcal{R}$

$$|R \cap S| \geq \varepsilon n \Rightarrow R \cap T \neq \emptyset$$

Informally, an ε -net is a small set of points that intersects every range of sufficiently large measure.

Theorem 4.40. (ε -Net Theorem) *There exists an ε -net $T \subseteq S$ such that*

$$|T| = \mathcal{O}\left(\frac{d}{\varepsilon} \log\left(\frac{d}{\varepsilon}\right)\right)$$

and in particular this bound does not depend on n .

Proof. (By Probabilistic Method) We will prove a weaker bound of $|T| = \mathcal{O}\left(\frac{\log n}{\varepsilon^2}\right)$. A more refined argument, using the VC dimension more directly, improves this to $|T| = \mathcal{O}\left(\frac{d}{\varepsilon} \log \frac{d}{\varepsilon}\right)$.

Choose points $x_1, \dots, x_m$ independently and uniformly from S , and let $T = \{x_1, \dots, x_m\}$

Fix $R \in \mathcal{R}$, denote p as the probability of the following

$$p = \frac{|R \cap S|}{n}$$

For each i , define a random variable $Z_i = \mathbb{1}_{\{x_i \in R\}}$. Then $Z_1, \dots, Z_m$ are independent Bernoulli random variables with mean p , so

$$E \left[\frac{1}{m} \sum_{i=1}^m Z_i \right] = p$$

A standard Chernoff bound shows that

$$\Pr \left[\bigcup_{i=1}^m Z_i = 1 \right] = \Pr \left[\frac{1}{m} \sum_{i=1}^m Z_i > 0 \right] \leq \Pr \left[\left| \frac{1}{m} \sum_{i=1}^m Z_i - p \right| > \varepsilon \right] \leq 2e^{-\varepsilon^2 m/2}$$

This would be considered as a bad event for a fixed R . Now, even though the family $\mathcal{R}$ may be infinite, when we only look at intersections with the finite set S , Sauer-Shelah tells us that there are at most n^d distinct sets of the form $R \cap S$. Therefore, by the union bound, the probability that there is some range R for which the sample behaves badly is at most

$$n^d \cdot 2e^{-\varepsilon^2 m/2}$$

If we choose

$$m = C \frac{\log n}{\varepsilon^2}$$

with C sufficiently large, then $n^d \cdot 2e^{-\varepsilon^2 m/2} < 1$. Hence there exists at least one choice of T for which all these bad events are avoided. $\square$

